Copyrighted Materials
Copyright © 2002 Oxford University Press Retrieved from www.knovel.com

1

Free Vibration of Systems with a Single Degree of Freedom

1.1 INTRODUCTION

In this chapter we'll examine the responses of systems with a single degree of freedom. This is one of the most important topics to master, since the more complicated cases (multi-degree-of-freedom and continuous systems) can often be treated as if they are simply collections of several, individual, single-degree-of-freedom systems. Once the responses of these simpler systems are well understood, branching out to the more complicated situations becomes relatively easy. Initially, we will ignore damping, since for many common structures the presence of damping changes the overall responses only slightly. Once we have a good grasp of how undamped problems behave, we'll look at the effects of damping.

Before diving in, it makes some sense to explain what "single degree of freedom" actually means. This idea is actually pretty simple—if an object is free to move in a particular direction, then it's got a degree of freedom in that direction. Thus, a bead that slides on a wire of an abacus has one degree of freedom. Figure 1.1 shows another example of a single-degree-of-freedom system, a mass in a channel. Since the mass is free to move only up or down the channel, it has one degree of freedom. We've used the coordinate x to describe the position of the mass. With this representation, it is clear

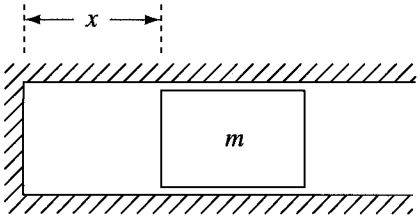


Figure 1.1 Single-degree-of-freedom system, single mass

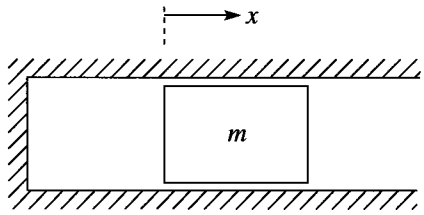


Figure 1.2 Single-degree-of-freedom system, single mass

what x stands for; it gives us the displacement of the mass from the left end. The more common way that we'll illustrate coordinates is shown in Figure 1.2. It looks likes the x coordinate is attached to the mass, but it really isn't. x motions are still referenced to the fixed channel, and the starting point of the x coordinate is aligned with the mass because the initial (zero displacement) reference occurs exactly at the mass's initial position. Any variations of the mass from the position it's currently in will produce a finite x reading.

Also, you should ignore any physical dimensions of the mass unless they're explicitly mentioned. For the purposes of this chapter we'll consider all masses to act as *point masses*; i.e., we can neglect their dimensions. Thus, although the masses are drawn in Figures 1.1 and 1.2 as having finite dimensions, that's really done just as an aid in visualizing them.

What about more than a single degree of freedom? A point mass on a flat plane has two degrees of freedom (it can move both up or down, and horizontally, for instance, as shown in Figure 1.3). A penny on a flat plane has three degrees of freedom—two to define the position of the penny's center and a third (an angular rotation) to describe what angle the penny is oriented in. Rigid bodies in three-space have six degrees of freedom—three translational (telling the location of the center of mass) and three rotational (telling the angular orientation of the body). In this text, you'll sometimes see *degree of freedom* written out but most often it will be abbreviated as DOF. Thus 2 DOF should be read as *two degrees of freedom.*

Conceptually, here's how the book will develop. We'll start by looking at SDOF, vibratory systems with no externally applied forces (Chapter 1). After we've gotten a feel for their characteristics, we'll add forcing and see how the systems behave (Chapters 2 and 3). This will give us an excellent grounding in the basic physics that all vibrational systems will display. Logically, we'll then move

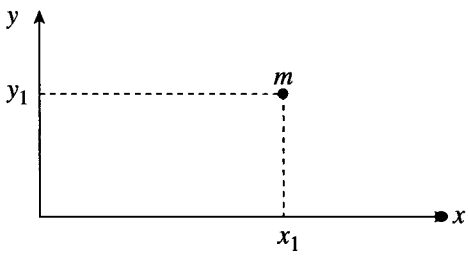


Figure 1.3 Two-degree-of-freedom system, single mass

on to multiple-DOF systems (Chapter 4). If we understand these systems, we've learned the most important part of the book, since essentially all modern vibrational analyses consider multiple-degree-of-freedom problems. As the number of degrees of freedom gets larger and larger, the system begins to look more like a continuum, rather than a collection of discrete pieces. Thus we'll spend some time looking at continuous systems and how they behave (Chapter 5). Following this, we'll develop some approximate solution methods that allow us to handle continuous problems that are just too difficult to solve exactly (Chapter 6). Finally, we'll show how some insight can help speed the solution procedure (Chapter 7) and touch on some of the aspects of modern modal analysis (Chapter 8).

1.2 TRANSLATIONAL VIBRATIONS—UNDAMPED

The first vibrational device that we'll discuss is a spring-mass system, something that you can make very easily. In fact, I strongly encourage you to take a minute right now and make one, since doing so will really help you in understanding several basic vibrations concepts. Although you can follow along without making an actual device, using one will greatly aid in developing your physical intuition.

In the good old days you'd have gotten a nifty spring included with the book. Sadly, too many people were ripping the springs out of the packaging and so the publisher (Oxford University Press) had to stop including them. Happily, they've agreed to supply the springs to the course instructor and, if all went well, he/she should already have them on hand. If not, ask him/her to give Oxford a buzz at PCG@OUP-USA.org and ask for them. If all else fails, you can try to use rubber bands instead of the spring.

Okay, assuming you've got a spring, the next thing to do is attach one end of it to a handy weight (a ceramic coffee cup will do nicely), and hold the free end of the spring in your hand. Make sure that the weight you've chosen isn't so heavy that the spring is stretched to near its limit, i.e., there must be a good bit of stretch remaining in the spring beyond its extension due to gravity. When you're done, you'll have a high-class vibrational test device like the one shown in this chapter's opener. (A cautionary note is needed here. If you decided to use rubber bands because you can't get hold of a spring, you should be aware that the rubber dissipates *much* more energy than a spring does, and so the vibrations will die away quickly and the forced tests to come in the next chapter will require more effort to make them work.)

These two elements, the mass and spring, are the most basic building blocks for vibrational analyses. Even though we know that our mass has some degree of internal flexibility, our model is considered to be infinitely rigid. Similarly, although the physical spring clearly has a mass of its own (it's not weightless, after all), our model will have none. Thus we are already approximating the real world. This idealization is necessary if we are to come up with reasonable mathematical models, but you should keep in mind that we are using approximations, which never exactly match reality. The fact that actual objects combine mass and stiffness (and damping) will become much more important when we look at the vibration of continuous systems, such as strings, rods, and beams. Happily, even though our system models will not *exactly* match reality, they can come very close to doing so, thus enabling us to use the approximations and feel confident that our results have physical meaning.

Having introduced masses and springs with words, it is appropriate to pin down exactly what we mean. Mass is a fundamental quantity, possessed by all macroscopic objects. This being a vibrations text, we won't get into the details of how physicists currently view matter, but instead use our well-known Newtonian view that mass times acceleration equals force and that the force generated by a mass in a gravitational field is called its weight.

Although we won't get any vibrations out of a mass by itself, it's useful to analyze the behavior of simple masses, especially since complicated objects such as airplanes and spacecraft can behave in a similar manner to a lumped mass under normal operating conditions. If we write out Newton's second law for a constant mass, we have

$$m\ddot{x} = f \tag{1.2.1}$$

Note that an overdot indicates differentiation with respect to time. If the applied force f is zero, then the solution to this equation is simply

$$x(t) = a + bt \tag{1.2.2}$$

where a and b are constants. Thus the mass can be stationary (if $b = 0$), or it can move with a constant speed b, in what we call rigid-body motion. In our case, there really isn't any body to speak of, since we're dealing with a mass particle. But more realistic extended bodies, such as aircraft, show the same behavior. If they aren't restrained, then they can remain in one spot or, depending on the initial conditions, translate and/or rotate at a constant speed. Those of you who've been in a jet when it hits a sudden downdraft can certainly attest that the whole plane can move sharply down without pitching or rolling to any great extent.

If we plot the relation between acceleration and force, we obtain Figure 1.4. The plot is linear and the slope varies with the mass m. Thus, the larger the mass, the less it accelerates for a given force.

Springs are a little more complicated than masses, but not by much. As we've already mentioned, to accelerate a given mass m at an acceleration a, we must apply a force f, where $f = ma$. In the case of a spring, an applied force doesn't produce an acceleration, but rather a displacement. Figure 1.5 shows the relationship between force and displacement for the illustrated spring: the graph

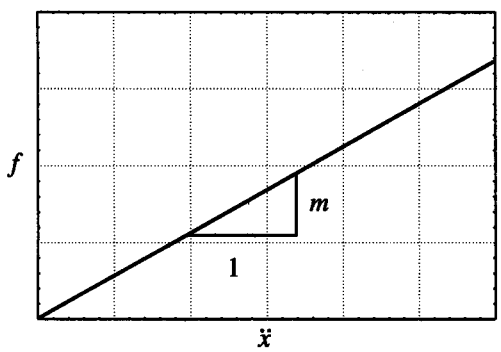


Figure 1.4 Force versus acceleration for a mass

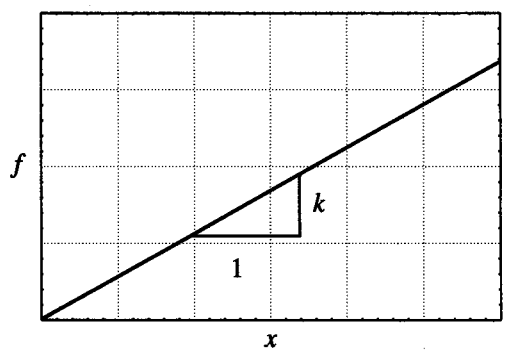


Figure 1.5 Force versus displacement for a spring

is linear, indicating that an increase in the applied force will produce a proportional change in the displacement. The constant of proportionality is commonly denoted by k and is called the spring constant. Thus

$$f = kx \tag{1.2.3}$$

When we increase k we're increasing the stiffness of the spring, making it harder to deflect.

Example 1.1

Problem A force deflection test yields the data shown in Figure 1.6. Determine the spring constant of the tested element.

Solution The slope of the data is equal to $\frac{40\,\text{N}}{5\,\text{mm}} = 8000\,\text{N/m}$. Thus the spring constant is equal to 8000 N/m.

Example 1.2

Problem Find the local spring constant for motions about $x = 1$ m for the nonlinear spring characteristic shown in Figure 1.7. The f axis is given in newtons and the x axis is in meters.

Solution Figure 1.7 shows a force versus deflection curve of a nonlinear spring for which

$$f(x) = x + x^3$$

To determine the linear spring characteristic, we have to examine the system for motions about a particular deflection, in this case $x = 1$ m. The local linear spring characteristic is given by

$$k = 1 + 3x^2\Big|_{x=1} = 4\,\text{N/m}$$

If the system is in static equilibrium when $x = 1$ m, then we now know that, for small oscillations about this equilibrium, the local spring constant will be 4 N/m.

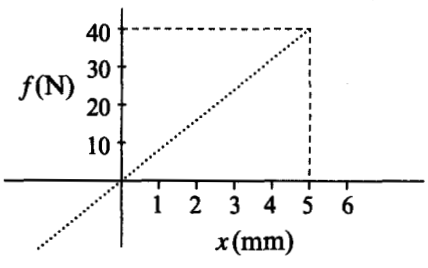


Figure 1.6 Experimental force/deflection data

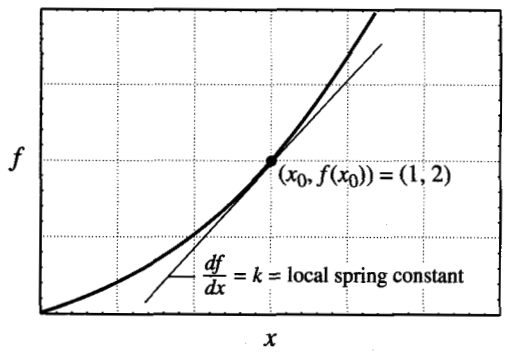


Figure 1.7 Local spring constant

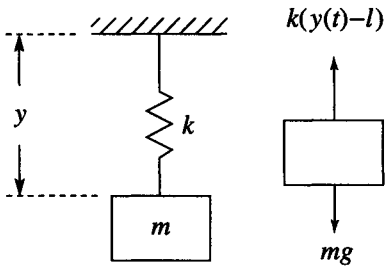


Figure 1.8 Free body diagram of a spring-mass system

Throughout the rest of the book we will use m_1, m_2, k_1, k_2, etc. to indicate both the particular object we're looking at and its associated physical quantity. Thus we might call a particular spring k_1 and also indicate its stiffness by k_1. Otherwise we'd have to say "consider the spring s_1, with spring constant k_1 and the mass n_1 with mass equal to m_1," which is long-winded and also uses up too many symbols. It'll be obvious from the sentence whether m_1 refers to the object itself or the mass of the object and whether k_1 refers to the spring or its spring constant.

The first experimental test we'll run will be a free vibration experiment. Keeping the free end of the spring fixed with the mass suspended beneath it, pull the mass down a bit and then release it. You'll note that the mass oscillates up and down at a fairly regular rate. Furthermore, it seems to oscillate with close to a constant amplitude. We can easily demonstrate that the mathematical equations of motion of this system exhibit these same properties. From the free body diagram of Figure 1.8 we see that the equation of motion for the system is

$$m\ddot{y}(t) = -k\big(y(t) - l\big) + mg \tag{1.2.4}$$

The spring's unstretched length is equal to l (thus the spring force is proportional to the stretch in the spring beyond this length) and g is the acceleration due to gravity. Although this representation is perfectly valid, it doesn't utilize an important fact, namely, that the oscillations will occur about the static equilibrium position of the mass due to gravity. This position is the one at which the restoring force due to the spring just balances the force due to gravity

$$k(y_0 - l) = mg \tag{1.2.5}$$

If we let $y(t) = y_0 + x(t)$, i.e., if x represents the motions about the equilibrium position, then we can rewrite (1.2.4) as follows:

$$m\ddot{x}(t) = -k(y_0 + x(t) - l) + mg \tag{1.2.6}$$

or, using our equilibrium equation (1.2.5) to eliminate mg,

$$m\ddot{x}(t) = -k(y_0 + x(t) - l) + k(y_0 - l) = -kx(t) \tag{1.2.7}$$

Moving all the dependent variables to the left side of the equation leaves us with

$$m\ddot{x}(t) + kx(t) = 0 \tag{1.2.8}$$

The difference between $y(t)$ and $x(t)$ is often a point of confusion, make sure that it is clear in your mind. Gravity simply serves to set up an equilibrium configuration for the system, and the system's vibrations occur about this equilibrium. You'll examine this more fully in some of the homework problems.

At this point, we'll stop showing the time dependency as (t) when it is clear from the context which of the variables are time dependent and which are just constants:

$$m\ddot{x} + kx = 0 \tag{1.2.9}$$

Equation (1.2.9) is an autonomous (no forcing), constant coefficient, linear differential equation. All constant coefficient, linear, ordinary differential equations have solutions of the form

$$x(t) = ae^{\lambda t} \tag{1.2.10}$$

That is, they exhibit exponential behavior. This is worth repeating. *ALL* autonomous, constant coefficient, linear differential equations have exponential solutions. Burn this into your brain; it underlies all the solution procedures we'll be meeting later in the book.

Using this assumed solution, (1.2.9) becomes:

$$(m\lambda^2 + k)ae^{\lambda t} = 0 \tag{1.2.11}$$

Since a can't be equal to zero (unless $x(t) = 0$ seems interesting to you) and $e^{\lambda t}$ isn't equal to zero for any finite time t, we're left with requiring that

$$m\lambda^2 + k = 0 \tag{1.2.12}$$

Therefore $\lambda^2 = -\frac{k}{m}$ or $\lambda_{1,2} = \pm i\sqrt{\frac{k}{m}}$ Thus the full solution can be written out as

$$x(t) = a_1 e^{it\sqrt{\frac{k}{m}}} + a_2 e^{-it\sqrt{\frac{k}{m}}} \tag{1.2.13}$$

Using the fact that $e^{\pm i\omega t} = \cos(\omega t) \pm i\sin(\omega t)$, we can just as easily express $x(t)$ as

$$x(t) = b_1 \cos\left(\sqrt{\frac{k}{m}}t\right) + b_2 \sin\left(\sqrt{\frac{k}{m}}t\right) \tag{1.2.14}$$

You can check that both (1.2.13) and (1.2.14) are valid solutions by using them in (1.2.9) and observing that they satisfy the equation of motion.

We see from (1.2.14) that $\sqrt{\frac{k}{m}}$ has the units of reciprocal seconds (since the argument of a sine or cosine is nondimensional) and thus has the dimensions of a frequency. We commonly refer to this particular frequency as the system's natural frequency, ω_n

$$\omega_n \equiv \sqrt{\frac{k}{m}} \tag{1.2.15}$$

We'll see when we start looking at forced vibrations that this frequency plays a very important role in a system's vibrational response.

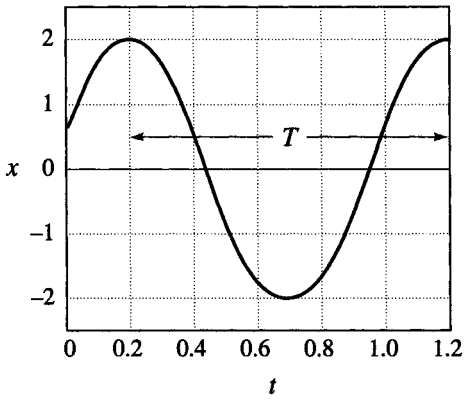


Figure 1.9 Sinusoidal waveform

If we plot (1.2.14) versus time for particular values of b_1, b_2, k, and m, we obtain Figure 1.9. You can see that the response is a periodic signal, meaning that the response repeats itself over and over. The time necessary for the signal to repeat (T) is the fundamental period of the oscillation, which is related to the natural frequency by

$$\omega_n = \frac{2\pi}{T} \tag{1.2.16}$$

In the absence of damping, the amplitude of the solution stays constant. Hopefully, when you performed the free vibration experiment with the spring and mass you saw the mass perform quite a few oscillations without much decay in the response. This was because the system didn't have much damping.

The details of what a particular response will look like depend upon the system's initial conditions. Given the right start, we could see a pure sine wave, a cosine, or a combination of the two. The next two examples will make this clear.

Example 1.3

Problem Given a sinusoidal signal with frequency equal to 1.0 rad/s and initial conditions $x_0 \equiv x(0) = 2$ and $\dot{x}_0 \equiv \dot{x}_0 = 0$, find the form of $x(t)$.

Solution From (1.2.14) we know

$$x(t) = b_1 \cos(t) + b_2 \sin(t)$$

Differentiating once yields

$$\dot{x}(t) = -b_1 \sin(t) + b_2 \cos(t)$$

Evaluating at $t = 0$ we have

$$x(0) = b_1 \cos(0) + b_2 \sin(0) = b_1 = 2$$
$$\dot{x}(0) = -b_1 \sin(0) + b_2 \cos(0) = b_2 = 0$$

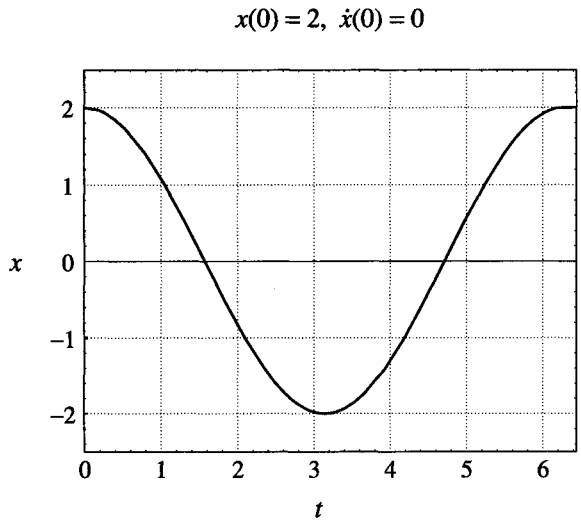


Figure 1.10 Free response of a spring-mass system

which gives us

$$x(t) = 2\cos(t) \tag{1.2.17}$$

This signal is shown in Figure 1.10.

Example 1.4

Problem Given a sinusoidal signal with frequency equal to 2π rad/s and initial conditions

$$x_0 = 1$$
$$\dot{x}_0 = 5$$

find the form of $x(t)$.

Solution From (1.2.14) we know

$$x(t) = b_1 \cos(2\pi t) + b_2 \sin(2\pi t)$$

and so, by differentiating, we have

$$\dot{x}(t) = 2\pi\,(-b_1 \sin(2\pi t) + b_2 \cos(2\pi t))$$

Thus

$$x(0) = b_1 \cos(0) + b_2 \sin(0) = b_1 = 1$$

$$\dot{x}(0) = 2\pi b_2 = 5 \quad \Rightarrow \quad b_2 = \frac{5}{2\pi}$$

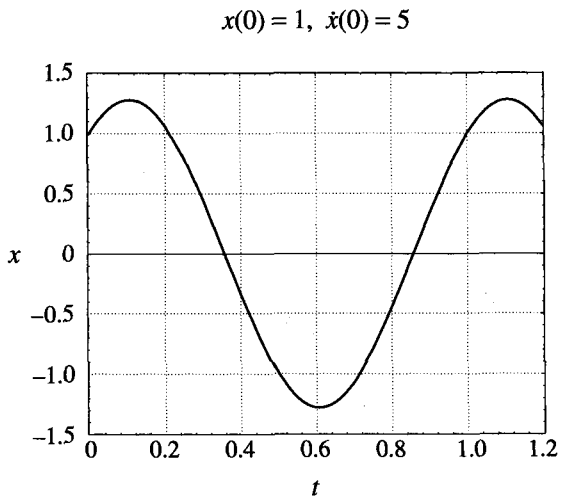


Figure 1.11 Free response of a spring-mass system

yielding

$$x(t) = \cos(2\pi t) + \frac{5}{2\pi}\sin(2\pi t) \tag{1.2.18}$$

This signal is shown in Figure 1.11.

Example 1.5

Problem A mass is suspended from a nonlinear spring. Find the equilibrium position of the mass and the natural frequency of small oscillations about this position. $m = 1$ kg and the force necessary to deflect the spring is given by

$$f = 1.5x + .2x^2$$

where x has the units of millimeters and f has the units of newtons.

Solution In an equilibrium condition, the force due to gravity is balanced by the force of the spring:

$$mg = 1.5x + .2x^2$$

Rewriting this (and using the known parameter values) yields

$$x^2 + 7.5x - 49.05 = 0$$

This quadratic equation has a positive root of $x = 4.194$ mm. k is the linearized spring constant of our nonlinear spring about the equilibrium position. Thus

$$k = \left.\frac{df}{dx}\right|_{x=4.194} = 1.5 + .4x\Big|_{x=4.194} = 3.178 \text{ N/mm} = 3178 \text{ N/m}$$

Although (1.2.17) and (1.2.18) are both perfectly valid ways to write the solutions, there are actually other forms that are just as correct. For instance, say we didn't like using cosines. Would

it be possible to express (1.2.17) in terms of just a sine? The answer is "yes." There's no problem with doing this. All we need to do is include the appropriate phase shift. Our original solution was $x(t) = 2\cos(t)$. If we want to use a sine, then we'd say $x(t) = 2\sin(t + \frac{\pi}{2})$. Check it out. At every time t, both these expressions produce the same answer. The $\frac{\pi}{2}$ term is called a *phase shift,* and it does what the name implies—it shifts the phase of the sine wave by $\frac{\pi}{2}$. This can also be viewed as shifting the effective origin of the time axis by $t = \frac{\pi}{2}$.

Since the idea of phase shifts will be of fundamental importance to understanding vibrational responses, we'll spend some time on the concept now. You'll recall from trigonometry that sines (or cosines) of the sum of two quantities can be expanded in terms of sines and cosines of those quantities through the formulas

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y) \tag{1.2.19}$$

$$\cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y) \tag{1.2.20}$$

Thus, if we wished to express

$$a_1 \sin(\omega_n t) + a_2 \cos(\omega_n t) \tag{1.2.21}$$

as a cosine with a phase shift ϕ we would say

$$d\cos(\omega_n t + \phi) = d\cos(\omega_n t)\cos(\phi) - d\sin(\omega_n t)\sin(\phi) \tag{1.2.22}$$

and, comparing the coefficients of $\cos(\omega t)$ and $\sin(\omega t)$ in (1.2.21) and (1.2.22), see that

$$d\cos(\phi) = a_2 \tag{1.2.23}$$

and

$$d\sin(\phi) = -a_1 \tag{1.2.24}$$

Squaring both sides of (1.2.23) and (1.2.24), adding them together and then taking the square root yields

$$d = \sqrt{a_1^2 + a_2^2} \tag{1.2.25}$$

and dividing $d\sin(\phi)$ by $d\cos(\phi)$ yields

$$\tan(\phi) = -\frac{a_1}{a_2} \tag{1.2.26}$$

or

$$\phi = -\tan^{-1}\left(\frac{a_1}{a_2}\right) \tag{1.2.27}$$

Thus we can always express the overall vibration as a cosine function with a phase shift and, in an exactly parallel manner, also express it as a sine function with a phase shift.

Example 1.6

Problem Put $2\sin(3t) + 4\cos(3t)$ into the form $d\cos(3t + \phi)$.

Solution For this problem, $a_1 = 2$ and $a_2 = 4$. Thus, using (1.2.25) we have

$$d = \sqrt{2^2 + 4^2} = \sqrt{20}$$

and (1.2.27) gives us

$$\phi = -\tan^{-1}\left(\frac{2}{4}\right) = -.464$$

Therefore

$$2\sin(3t) + 4\cos(3t) = \sqrt{20}\cos(3t - .464)$$

Figure 1.12 shows what the final waveform looks like and shows how the phase shift acts to move the cosine wave over to the right. The effective time origin of the cosine wave is at $t = \frac{.464}{3} = .155$.

Okay, now we know how to find the free response of a system that consists of one mass and one spring. But what if our system is more complex than this? Suppose it involves several springs. What do we do? Well, let's consider the problem.

Say we have a mass supported by two springs in *series*, as shown in Figure 1.13*a*. This is a very common situation. For instance, if you were modeling an automobile, you'd want to take account of the spring stiffness of the tire first (our k_1). Next, you'd run into the actual suspension springs of the vehicle (our k_2). Finally, you'd have the mass of the car body (our m).

How can we determine the equivalent k (Figure 1.13*b*) for this system? Well, the approach we'll use is one that you'll find helpful in the future when you need to determine the boundary conditions for continuous problems (and, of course, it's helpful here as well). What we'll do is create a new problem, shown in Figure 1.14*a*. For this problem, we've taken the two springs that are in series and

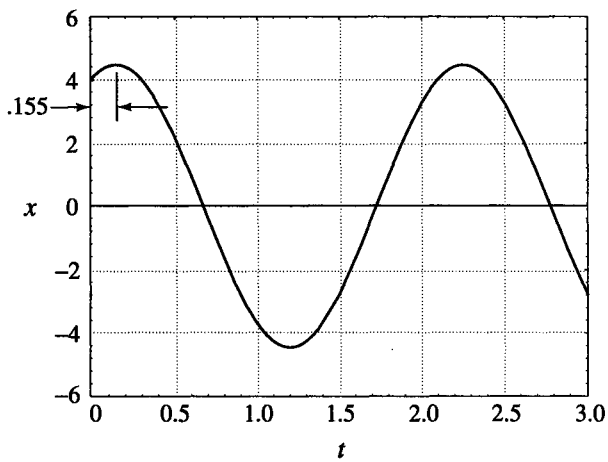


Figure 1.12 $4.47\cos(3t - .464)$

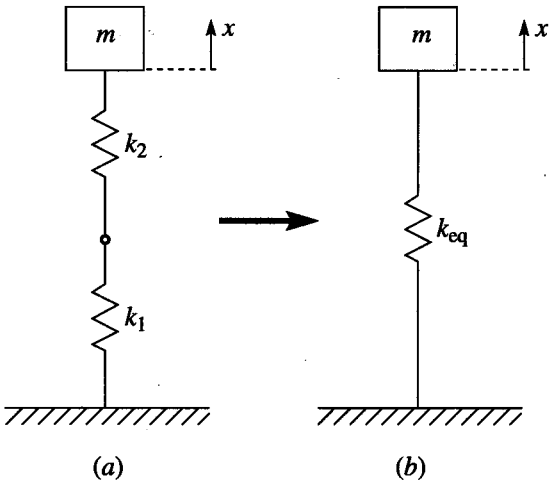


Figure 1.13 Representing two springs in series as a single equivalent spring

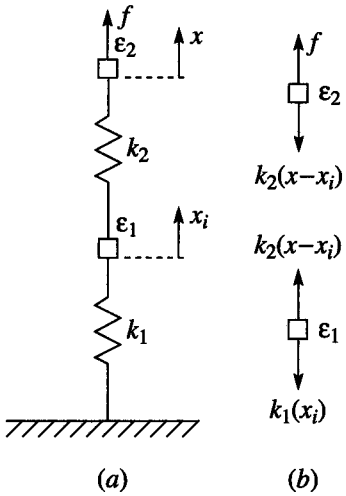


Figure 1.14 Free-body diagram with two springs in series

put a tiny mass ε_1 between them and also put another tiny mass ε_2 at the end of k_2. Finally, we've applied a force f to the top mass.

As indicated in Figure 1.13, what we eventually want is an expression like

$$f = k_{eq}x \tag{1.2.28}$$

where k_{eq} is our equivalent spring constant. The reason for introducing the tiny masses is to make it easier to apply Newton's laws to this problem in a way that you're familiar with. However, once we get the equations of motion, we'll set ε_1 and ε_2 to zero, thus reducing the problem to our original one. So the ε's are just a conceptual aid.

Going ahead and determining the equations of motion for this system (using the free body diagrams from Figure 1.14*b* gives us

$$\varepsilon_2\ddot{x} = -k_2(x - x_i) + f \tag{1.2.29}$$

and

$$\varepsilon_1\ddot{x}_i = k_2(x - x_i) - k_1x_i \tag{1.2.30}$$

Note that no gravitational force (mg) appears in the free body diagram because, as mentioned previously, we're concerned with vibrations about the system's static equilibrium position.

We'll now set the ε's to zero, thus giving us the correct static equations

$$k_2(x - x_i) = f \tag{1.2.31}$$

$$k_2(x - x_i) - k_1x_i = 0 \tag{1.2.32}$$

If we use (1.2.31) to eliminate $k_2(x - x_i)$ from (1.2.32), we'll obtain

$$f = k_1 x_i \tag{1.2.33}$$

or

$$x_i = \frac{f}{k_1} \tag{1.2.34}$$

Substituting this value for x_i into (1.2.32) gives us

$$k_2\left(x - \frac{f}{k_1}\right) - f = 0 \tag{1.2.35}$$

which, when rearranged, yields

$$\left(\frac{k_1 k_2}{k_1 + k_2}\right) x = f \tag{1.2.36}$$

Comparing (1.2.36) with (1.2.28), we see that we've found our equivalent spring constant:

$$k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2} \tag{1.2.37}$$

If you've taken a circuits course, this will remind you strongly of the formula for the equivalent resistance of two resistors in parallel, *not* two resistors in series. This is a point that sometimes causes confusion, since the schematic representation of two springs in series looks very much like two resistors in series (Figure 1.15). The relevant equations for this electrical example are:

$$v - v_i = r_2 i \tag{1.2.38}$$

and

$$v_i = r_1 i \tag{1.2.39}$$

Combining (1.2.38) and (1.2.39) produces

$$v = (r_1 + r_2) i \tag{1.2.40}$$

yielding an equivalent resistance of $r_{\text{eq}} = r_1 + r_2$.

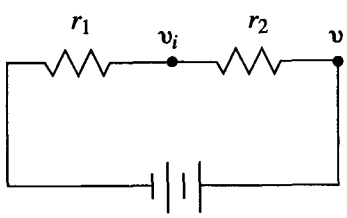


Figure 1.15 Electrical resistors in series

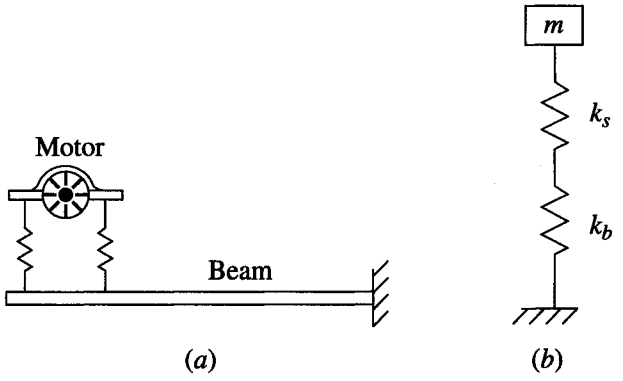


Figure 1.16 Motor on an elastic support

Example 1.7

Problem The system shown in Figure 1.16*a* is a motor, m, with a suspension, k_s. The motor is attached to a flexible support beam, with stiffness k_b. The simplified model is shown in Figure 1.16*b*. Find the natural frequency of the combined system. $m = 20$ kg, $k_s = 1000$ N/m, and $k_b = 200$ N/m.

Solution From (1.2.37) we have

$$k_{eq} = \frac{k_s k_b}{k_s + k_b} = 166.6 \text{ N/m}$$

$$\omega_n = \sqrt{\frac{166.6}{20}} = 2.887 \text{ rad/s}$$

Calculating the equivalent spring constant for springs in parallel is easier than the series arrangement. Figure 1.17 shows this case and the equivalent formulation we're trying for. Figure 1.18 illustrates the relatively straightforward free body diagram for this problem. Applying $f = ma$

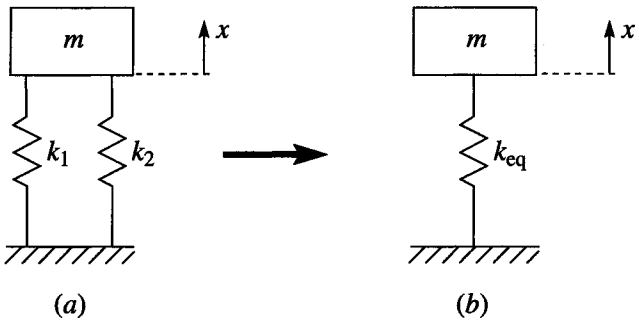


Figure 1.17 Representing two springs in parallel as single equivalent spring

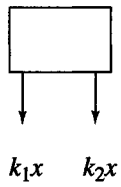


Figure 1.18 Free-body diagram for a mass with two springs in parallel

in the vertical direction gives us

$$m\ddot{x} = -k_1 x - k_2 x \tag{1.2.41}$$

or

$$m\ddot{x} + (k_1 + k_2)x = 0 \tag{1.2.42}$$

Thus the equivalent spring constant for two parallel springs is given by

$$k_{eq} = k_1 + k_2 \tag{1.2.43}$$

Note that this looks just like the electrical resistance formula we derived a moment ago. If you look at the equivalent resistance of resistors in parallel, you'll have the completed picture—that series arrangements of springs are like parallel arrangements of resistors and parallel arrangements of springs are like series arrangements of resistors. There actually is a logical way to look at both systems in a way that predicts this behavior, but we won't go into it here. The interested reader can check [Reference 8] for more information. Now that we've got (1.2.37) and (1.2.43), we can determine the equivalent spring constant for any particular arrangement of springs simply by breaking the more complicated problem down into smaller pieces.

Example 1.8

Problem What is k_{eq} for the system illustrated in Figure 1.19*a*?

Solution For k_{eq_1} (Figure 1.19*b*) we have

$$k_{eq_1} = \frac{k_2 k_2}{k_2 + k_2} = \frac{k_2}{2}$$

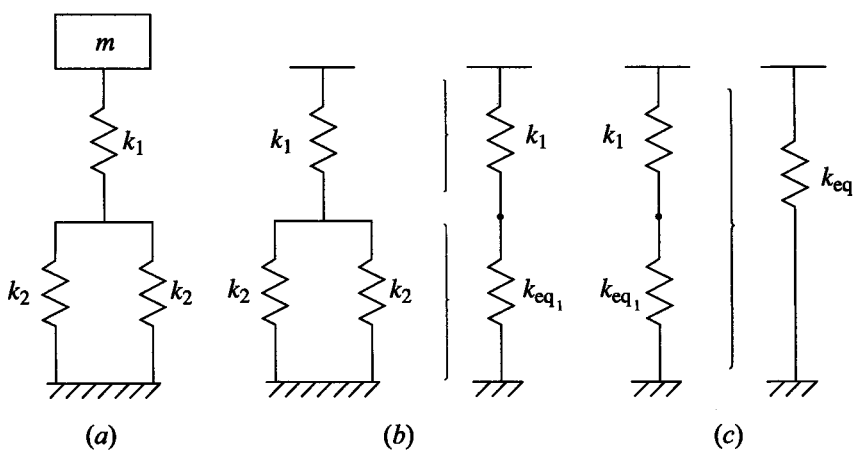


Figure 1.19 Mass on a multi-spring foundation

and for k_{eq} (Figure 1.19c) it's

$$k_{eq} = k_{eq_1} + k_1 = \frac{k_2}{2} + k_1$$

Even though we really never have pure springs and masses in practice, we often approximate our systems as if the lumped elements only exhibited a single characteristic. For instance, examine Figure 1.20, which shows an extensible bar with a lumped mass at the end. We know that the bar itself has mass, but if the lumped mass is much greater than the bar's mass, we might decide to model the bar as a pure spring. We'll revisit this problem in Chapter 7, to see how more exact solutions compare to our approximate ones.

If we're going to ignore the bar's own mass, then the system can be represented as a lumped mass m_1 at the end of a spring. All we need to do is determine the spring constant, and then we can solve for the system's natural frequency. The governing equation static for this case is

$$EA\xi_x = f \tag{1.2.44}$$

where f is a constant force applied at the end of the bar and EA is the bar's stiffness. Integrating this gives us

$$EA\xi(x) = fx + b \tag{1.2.45}$$

Our boundary condition of zero displacement at $x = 0$ causes b to equal zero and, evaluating this expression at $x = l$, we find

$$EA\xi(l) = fl \tag{1.2.46}$$

This can be rewritten as

$$f = \frac{EA}{l}\xi(l) \tag{1.2.47}$$

If we draw a parallel with our well-known spring force equation ($f = kx$), we can identify $\frac{EA}{l}$ as our equivalent spring constant k_b. Thus our natural frequency is given by

$$\omega_n = \sqrt{\frac{k_b}{m_1}} = \sqrt{\frac{EA}{lm_1}} \tag{1.2.48}$$

As we've mentioned, the accuracy of this approximation depends upon the end mass m_1 being substantially greater than the bar's mass. To see this, we can look at two cases. First, we'll consider the

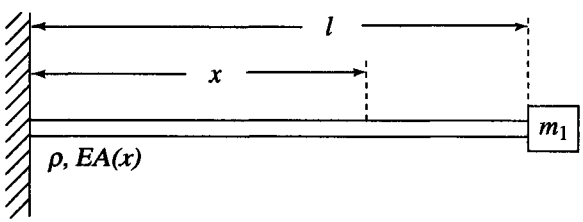


Figure 1.20 Schematic of a bar with a lumped mass

case for which the bar's density ρ is 1 kg/m. The other parameters will be $l = 1$ m, $m_1 = 10$ kg, and $EA = 10$ N. Thus the total mass of the bar is one-tenth that of the end mass. For these values we'll obtain $\omega_n = 1.00$ rad/s. If we were to solve this problem exactly (using the techniques of Chapter 5), we'd find that the exact answer is .9836 rad/s. Thus our error is only around 2%.

If we now increase the bar's density by a factor of 10, to 10 kg/m, our estimate doesn't change at all (since it ignores the bar's mass). However the mass certainly affects the actual answer, and lowers the natural frequency to $\omega_n = .8603$ rad/s, giving us an error of 16%. This isn't really too bad if you consider that the bar's mass is now equal to the end mass. It's actually possible to correct our approximation a good deal by using some engineering insight, and we'll go into this in Chapter 7.

As an aid to the homework problems, Appendix B contains lumped spring constants for a variety of common situations. In addition, Appendix C contains physical constants for several common structural materials.

1.3 ROTATIONAL VIBRATIONS AND LINEARIZATION

Although many systems involve only translational motion of the sort we've been studying, many also include rotational motions. As long as the rotation angles are small, these systems can be analyzed just like the systems we've already seen. Consider, for example, a simple pendulum (Figure 1.21). This consists of a point mass, suspended from a massless rod. The generalized coordinate is θ, the angle by which the rod deviates from the vertical position.

It's important to realize that there are actually two equilibrium positions for this problem. The obvious one is the one shown, for which θ is zero. A pendulum is quite happy to remain near this position, oscillating back and forth about it. However, there is another equilibrium, namely, when θ is 180 degrees. In this case the pendulum points directly upward. This is what we call an unstable equilibrium, since it is precarious at best. If we very, very carefully place the pendulum in this upright position, it should remain upright. At least that's what the mathematics will tell us. But the mathematics will also tell us that any disturbance, no matter how small, will cause the pendulum to fall away from this equilibrium (and end up moving about the stable equilibrium at $\theta = 0$). Thus in this and other examples, you'll want to be sure you're analyzing the system about the appropriate equilibrium. We'll mention how to do this after we've found the equations of motion.

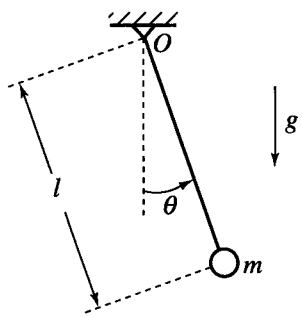


Figure 1.21 Simple pendulum

Constructing a free body diagram of this system and summing moments about the attachment point 0 leads to

$$\sum \text{moments}_0 = -mgl\sin(\theta) \tag{1.3.1}$$

This is equal to the time rate of change of the angular momentum about O,

$$ml^2\ddot{\theta} = -mgl\sin(\theta) \tag{1.3.2}$$

or

$$ml^2\ddot{\theta} + mgl\sin(\theta) = 0 \tag{1.3.3}$$

Keep in mind that in this example the point O is stationary, and so the equation used is correct. If the attachment were moving, we'd have additional terms in our moment equation.

All right, at this point we'll look at equilibria. Simply put, an equilibrium position is one for which the system is happy to remain at rest. No motion. Not even a little. A straightforward way to find the equilibria is to take the equations of motion ((1.3.3) in our case) and just cross off the terms that imply motion. For us, this means crossing off the $\ddot{\theta}$ term. Obviously, if there's no motion, then all derivatives with respect to time will have to be zero. This leaves us with

$$mgl\sin(\theta) = 0 \tag{1.3.4}$$

Since the mgl term isn't zero, we need to find the point at which $\sin(\theta)$ is zero. This occurs at $\theta = 0$ and at $\theta = \pi$. Thus our two equilibrium positions involve the pendulum hanging straight down or standing straight up, as mentioned a second ago.

We can't solve (1.3.3) in the same manner we used for the translational problems earlier because this equation is nonlinear (i.e., the equation doesn't just involve the first power of the dependent variable θ and its time derivatives). To solve this problem, we need to *linearize* the equation. Simply put, this means that we expand out all terms involving the dependent variables and their time derivatives and then throw away all that don't consist solely of the dependent variables (and their time derivatives) raised to the first power. Thus, any terms like $\theta\dot{\theta}$, θ^2, and $\theta^2\ddot{\theta}$ will be discarded, while terms like θ, $\dot{\theta}$, and $\ddot{\theta}$ will be kept. When equations involve only terms like θ, $\dot{\theta}$, and $\ddot{\theta}$ (dependent variables and their derivatives raised to the first power), we call them linear. If other terms, such as θ^2, $\dot{\theta}\theta$, and θ^3, are present, then the equation is nonlinear.

Since we know that $\sin(\theta)$ can be expanded in a power series:

$$\sin(\theta) = \theta - \frac{1}{3!}\theta^3 + \cdots \tag{1.3.5}$$

we can discard everything except θ to obtain

$$\ddot{\theta} + \frac{g}{l}\theta = 0 \tag{1.3.6}$$

We can safely discard the higher order terms because they become vanishingly small as θ is made small; i.e., if θ is small, θ^2 is *really* small—small enough to ignore.

Note that we've divided through by ml^2. Also note that when we expanded out, we were doing the expansion about a particular equilibrium position (in this case $\theta = 0$). This is always the case. You get your nonlinear equations of motion, find the equilibria, and then linearize about a particular equilibrium.

Equation (1.3.6) is linear and can be solved in the same manner as the translational problems. Since we've thrown away quite a few terms to get to this linearized form, we need to ask whether the resulting equation has any validity. Clearly, tossing out the terms as we did will be sensible only if, during the motion of the actual system, these terms are negligibly small. This condition is equivalent to saying that $\sin(\theta)$ is very close to equaling θ itself. We know from trigonometry that this is true only if θ is small. How small? You can easily verify with your calculator that, if θ is less than .24 radian, then the error in approximating $\sin(\theta)$ by θ is less than 1% and the error for angles up to .53 radian is under 5%. These are pretty fair-sized angles, and so we see that the approximation is a good one for realistic amplitudes of oscillation.

Example 1.9

Problem How long must a simple pendulum (lumped mass on the end of a massless rod) be for the period of its oscillation to equal 1 second?

Solution From (1.3.6) we have

$$\omega_n^2 = \frac{g}{l}$$

If the period is 1 second long, then

$$\omega_n = \frac{2\pi}{T} = 2\pi$$

Thus

$$(2\pi)^2 = \frac{9.81}{l} \quad \Rightarrow \quad l = .248\,\text{m}$$

Example 1.10

Problem Find the natural frequency of the streetlight shown in Figure 1.22*a*, taking gravity into account. Compare the result to the case in which gravity is neglected. View the pole as being infinitely rigid. The mass of the light is the dominant mass and is represented by m (shown in Figure 1.22*b*). The rotational stiffness of the arm is represented by a lumped rotational spring, k_θ. The values of the various parameters are given by $m = 50$ kg, $g = 9.81$ m/s^2, $k_\theta = 29,393$ N·m, and $l = 3$ m.

Solution When the arm is horizontal, the spring is unstretched. Gravity is going to work to push the end mass down while the torsional spring will resist this motion. Thus the equilibrium position will be somewhere between $\theta = 0$ and $\theta = \frac{\pi}{2}$. If we sum moments about 0, we'll find

$$ml^2\ddot{\theta} = -k_\theta\theta + mgl\cos(\theta)$$

Setting the time derivatives to zero gives us

$$-k_\theta\theta + mgl\cos(\theta) = 0$$

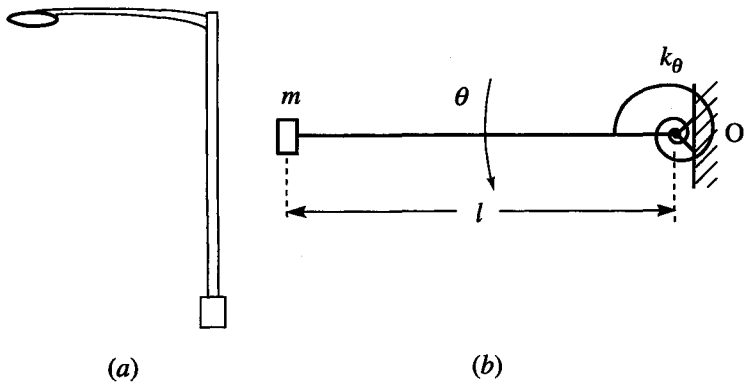


Figure 1.22 Streetlight and lumped approximation

or

$$k_\theta \theta = mgl\cos(\theta)$$

Using the given parameter values, this can be solved to give

$$\theta_{\text{eq}} = .05 \text{ rad}$$

If we expand about the equilibrium, θ_{eq}, by representing θ as

$$\theta = .05 + \eta$$

(where η is small), we'll have

$$ml^2 \frac{d^2}{dt^2}(.05 + \eta) + k_\theta(.05 + \eta) - mgl\cos(.05 + \eta) = 0$$

We now need to expand $\cos(.05 + \eta)$ into $\cos(.05)\cos(\eta) - \sin(.05)\sin(\eta)$. Since η is assumed to be small, this can be approximated by $\cos(.05) - \sin(.05)\eta$, (where we've used $\sin(\eta) \approx \eta$ and $\cos(\eta) \approx 1$).

Substituting this approximation into our equation of motion gives us

$$ml^2\ddot{\eta} + k_\theta(.05 + \eta) - mgl\Big(\cos(.05) - \sin(.05)\eta\Big) = 0$$

Expanded and simplified, this will finally yield

$$ml^2\ddot{\eta} + \Big(k_\theta + mgl\sin(.05)\Big)\eta = 0$$

If we go ahead and substitute in the appropriate values for m, l, and so on, we'll have

$$450\ddot{\eta} + 29,467\eta = 0$$

giving us a value of 8.09 rad/s as the system's natural frequency. Note that in the absence of gravity, the natural frequency would be $\sqrt{\frac{29,393}{450}} = 8.08$ rad/s. Thus the effect of the slight steady state rotation due to gravity had a negligible effect of the system's natural frequency.

Example 1.11

Problem Find the equations of motion for the system shown in Figure 1.23. Neglect gravity.

Solution The system we'll be examining contains both rotational and translational components. The horizontal bar can rotate about its hinge and the angle of rotation is given by θ. Rotations are resisted by the springs k_1 and k_2. As in the preceding examples, we'll assume that all rotations are small, and thus the displacement of the right end is given by $l_2\theta$. A mass is connected by a spring to the right end of the bar and an external vertical force $f(t)$ is applied to the same end.

The first step in solving this problem is to construct a free body diagram that shows what forces are acting on the system (Figure 1.24). Note that no reaction forces are drawn for the left end of the bar because the bar is physically pivoting about that point and thus the reaction forces will apply no moments there.

Summing moments about the pivot gives us

$$I_A\ddot{\theta} = -k_1(l_1\theta)l_1 - k_2(l_2\theta - x)l_2 + l_2 f(t)$$

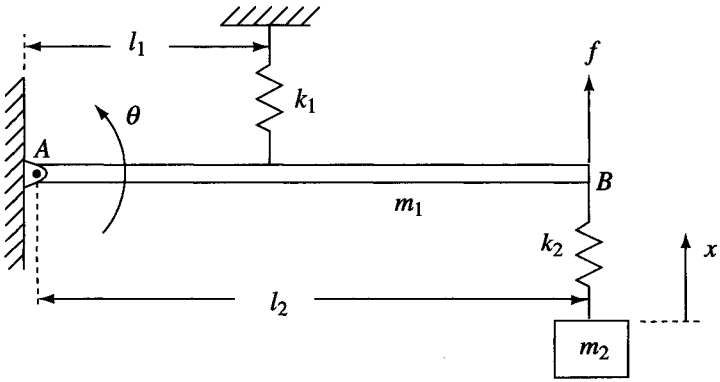


Figure 1.23 Spring-restrained/spring-mass system

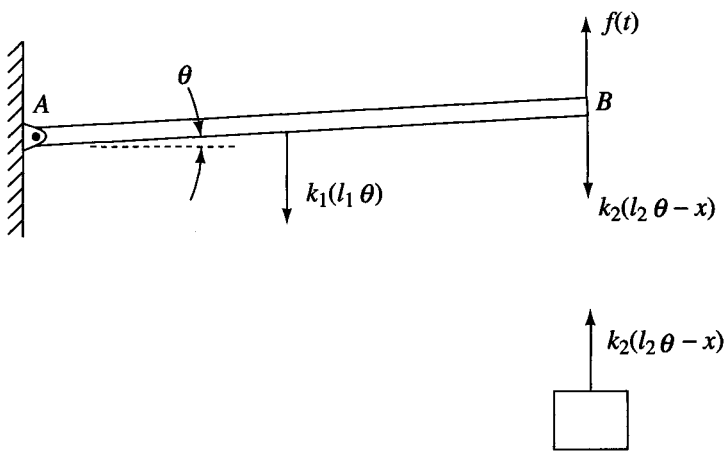


Figure 1.24 Free body diagram for spring-restrained/spring-mass system

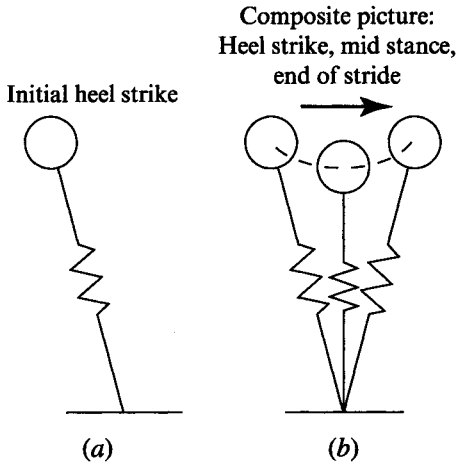


Figure 1.25 Spring-mass model of a walking organism

while equating force and acceleration in the vertical direction for the lumped mass yields

$$m_2\ddot{x} = k_2(l_2\theta - x)$$

Putting all the dependent variables on the left-hand side then gives us the final equations of motion

$$I_A\ddot{\theta} + (k_1 l_1^2 + k_2 l_2^2)\theta - k_2 l_2 x = l_2 f(t)$$

$$m_2\ddot{x} + k_2 x - k_2 l_2 \theta = 0$$

Finally, before moving on to the topic of damping, take a look at Figure 1.25. The illustrated model is used in state-of-the-art analyses of locomotion [4] and applies to creatures as diverse as crabs, cockroaches, and humans. As you can see, the model is a single-mass, single-spring system, exactly the same sort of system as we've been studying. The mass represents the creature's lumped mass and the spring represents the leg's stiffness. Figure 1.25*a* shows the system as the foot is just touching down and Figure 1.25*b* shows three snapshots that cover the full range of ground contact positions, from initial contact to the instant at which contact is lost.

The point of showing this example is to emphasize that even though you're not far into the book yet, you've already encountered models that are actually being used in the real world, and you have the knowledge to do something useful with these models.

1.4 VISCOUS DAMPING

The difficulty with the analyses so far is that our systems will never stop moving. They'll oscillate forever, something that clearly doesn't happen in the real world. When we pull down the coffee cup in our spring-mass experiment and then release it, the magnitude of the resulting oscillations becomes less and less as time goes on. The energy of the system is slowly dissipated away. Can we modify our equations so that they demonstrate this behavior? We've basically got two ways to go if we wish to

add dissipation (damping) to our system model. First, we can try and accurately model the processes by which dissipation occurs in real life and add the appropriate terms to our equation of motion. In real life, damping is a very intricate and involved process, and to correctly model it we would need a complex set of equations that are very system dependent.

The other approach would be to ignore the "real" damping behavior and try and convince ourselves that a simple linear term is a good enough approximation. The advantages to this approach are that it's *much* simpler than doing it the "correct" way, and we can obtain closed-form solutions for our problems rather than having to resort to numerical (computer-based) solutions.

Given these two choices, the scientist in us would opt for the first approach, while our engineering half would probably vote for the simple solution. Since this is an engineering text, the second approach wins! Before those of you with a more scientific bent cry foul, you should know that the simple linear approach, although not exactly correct, is usually very close. When the level of dissipation is small, as it almost always is for engineering structures, it really doesn't make a great deal of difference in the final results if we use a linear damping approximation or a more realistic one. Only when the damping is large do we find sizable discrepancies creeping in between the different approaches. We'll see in a moment what is really meant by "large" and "small" values of damping.

Thus far, our formulations have involved terms for which the force is proportional to acceleration ($m\ddot{x}$) and displacement (kx). What we'll do now is add one that's proportional to velocity

$$f = c\dot{x} \tag{1.4.1}$$

A damper that exhibits a linear characteristic like this is often referred to as a *viscous damper*. It makes physical sense to let our damping term scale with velocity, since we're very familiar with the fact that drag on cars and boats increases with velocity, and thus the damping force in our equations should increase with velocity. (By the way, the drag on cars actually doesn't increase linearly; it scales more closely with the square of the velocity.) Besides being called viscous, you'll also see this type of damping referred to as *linear damping*.

Putting this new term into our equation of motion yields

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{1.4.2}$$

Although we threw this damping term in, we could just as well have postulated the existence of a physical damper (shown in Figure 1.26) that is attached in parallel with the spring of our system. If we assume that the force generated is proportional to velocity, then a free body force balance would

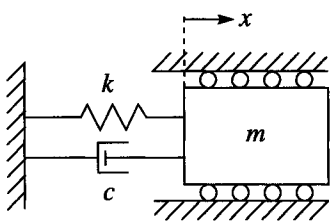


Figure 1.26 Spring-mass damper system

give us

$$m\ddot{x} = -c\dot{x} - kx \tag{1.4.3}$$

which is simply (1.4.2) arranged in a different order. The nice thing about visualizing the damper as a physical component is that it allows us to see why the damper will act to reduce the motions in the system. Since the force is directed toward the system's equilibrium position (just as the spring's force is) and is proportional to velocity, it generates a force that's directed toward the equilibrium point when the mass is moving away from that point. The faster the mass is moving, the greater the force acting on it.

Having added what we think will give us a damping effect, the next step is to solve the equations and see what we get. To start, we'll deal with the unforced case, just as we did for the undamped case. Since our equation is still linear, and remembering that all unforced, linear, constant coefficient, ordinary differential equations have exponential solutions, we'll solve the problem in exactly the same manner as we did for the undamped case, by assuming a solution of the form

$$x = ae^{\lambda t} \tag{1.4.4}$$

Using (1.4.4) in (1.4.2) we obtain

$$m\lambda^2 ae^{\lambda t} + c\lambda ae^{\lambda t} + kae^{\lambda t} = 0 \tag{1.4.5}$$

Grouping terms gives us

$$ae^{\lambda t}(m\lambda^2 + c\lambda + k) = 0 \tag{1.4.6}$$

Once again, $e^{\lambda t}$ is never zero for finite t and $a = 0$ is uninteresting, so we are left with

$$m\lambda^2 + c\lambda + k = 0 \tag{1.4.7}$$

Solving this gives us

$$\lambda_{1,2} = \frac{1}{2m}(-c \pm \sqrt{c^2 - 4mk}) \tag{1.4.8}$$

Although we could leave the solution in this form, we've already seen that dividing our undamped equation by m led us to the natural frequency ω_n and put the equation into a very neat form. If we carry through the division by m as in (1.4.8), we'll obtain

$$\lambda_{1,2} = -\frac{c}{2m} \pm \sqrt{\frac{1}{4}\left(\frac{c}{m}\right)^2 - \frac{k}{m}} = -\frac{c}{2m} \pm \sqrt{\frac{1}{4}\left(\frac{c}{m}\right)^2 - \omega_n^2} \tag{1.4.9}$$

Note that all we really need to do now is specify the ratio of c to m. The absolute values of these quantities don't greatly matter, which is one of the facts that motivated our division by m in the first place. Although it isn't clear why at the moment, it's convenient to replace $\frac{c}{m}$ with $2\zeta\omega_n$. If you'll suspend your disbelief for just a second, you'll see why.

Using this definition, our equation of motion becomes (after dividing by m)

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0 \tag{1.4.10}$$

Substituting $2\zeta\omega_n$ for $\frac{c}{m}$ in (1.4.9), we obtain

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{1.4.11}$$

This form is great as long as ζ is larger than one. But for $\zeta < 1$ we have the square root of a negative number. In this case we'll get an imaginary value, and the roots are most conveniently expressed as

$$\lambda_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2} \tag{1.4.12}$$

Note that we get two forms that depend upon ζ's value. For (1.4.11), both roots are real, negative numbers. This means that the overall solution to the free vibration problem will include two decaying exponentials. A typical response to an initial displacement is shown in Figure 1.27. For this example ω_n is 6.28 and ζ is 2. The mass starts off displaced from the origin and then smoothly approaches it as time increases.

This is different from the behavior we'll find for the kinds of root expressed by (1.4.12). For this case we no longer have two negative, real roots but, rather, complex roots. The real part of each root is the same, while the imaginary part differs in sign. Roots like these are called *complex conjugates* and are very common solutions for vibrations problems. To see what sort of overall solutions these roots support, we'll use the following trigonometric identity:

$$e^{ix} = \cos(x) + i\sin(x) \tag{1.4.13}$$

Thus, instead of solutions that look like

$$x(t) = a_1 e^{-\zeta\omega_n t} e^{i\omega_n t\sqrt{1-\zeta^2}} + a_2 e^{-\zeta\omega_n t} e^{-i\omega_n t\sqrt{1-\zeta^2}} \tag{1.4.14}$$

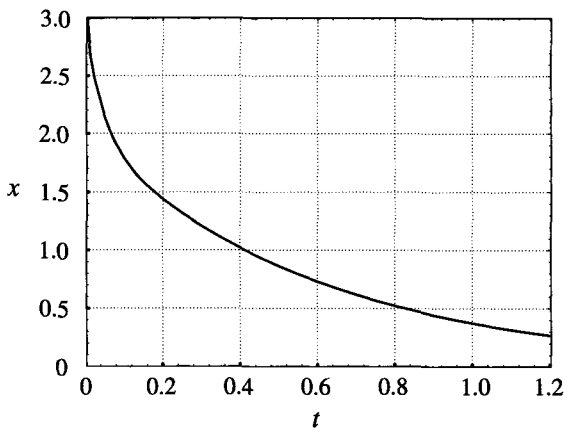


Figure 1.27 Overdamped response

we can express the solution as

$$x(t) = e^{-\zeta\omega_n t}\,[b_1\cos(\omega_d t) + b_2\sin(\omega_d t)] \tag{1.4.15}$$

where $\omega_d \equiv \omega_n\sqrt{1-\zeta^2}$.

Thus we have a solution made up of a sinusoidal part multiplied by a decaying exponential. ω_d is known as the *damped natural frequency*. Since the exponential is equal to 1.0 for $t = 0$ and decays from there, the effect of multiplying the sine or cosine by it is to cause the oscillations to decay within an exponential envelope. This is shown graphically in Figure 1.28 for the response

$$x(t) = e^{-1.256t}\cos(6.153t) \tag{1.4.16}$$

which corresponds to $\omega_n = 6.28$ and $\zeta = .2$.

Note that the general form of this type of solution is $x(t) = e^{-c_1 t}\cos(c_2 t + \phi)$. If changes to our system cause c_1 to increase, then the speed at which the oscillations decay is increased. Also, increasing c_2 causes the frequency of the oscillations to increase. Since we can control either of these by varying ω_n and ζ, we have the tools necessary for modifying the decay characteristics of our system.

The critical parameter that divides the responses between oscillatory ones (1.4.12) and purely decaying ones (1.4.11) is ζ. For $\zeta > 1$ we'll have exponential decay (because the number under the square root of (1.4.11) is positive) and for $\zeta < 1$ we'll have decaying oscillations. This is the main reason for the introduction of ζ; it characterizes these two response regimes. Different books call ζ different things (damping coefficient, damping ratio, damping factor). In this book we'll refer to it as the damping factor. If our system has a ζ that's less than 1.0, we call the system *underdamped*. Systems with ζ's greater than 1.0 are *overdamped*, while systems having ζ's of exactly 1.0 are *critically damped*.

Looking at our definition of ω_d ($\omega_d = \omega_n\sqrt{1-\zeta^2}$), we see that the frequency of our damped oscillations depends upon both ω_n and ζ. For small values of ζ (lightly damped) the oscillation rate is essentially ω_n. Increasing the damping factor serves to reduce the oscillation rate. This makes physical

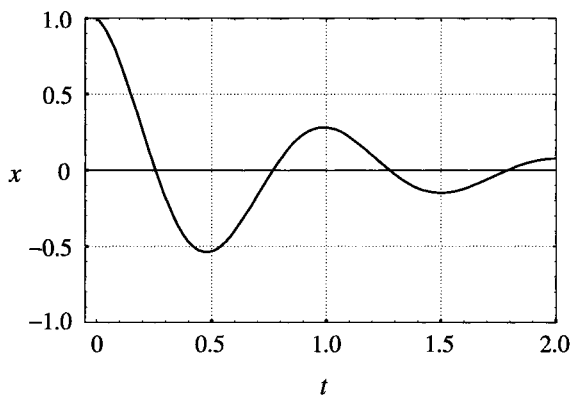


Figure 1.28 Underdamped response

sense, since the mass will have more difficulty in moving if the damping is raised, thus lowering the overall rate of oscillation.

Although ζ is the damping factor for our system, keep in mind that it is *not* the sole determiner of the response's decay rate. Equation (1.4.15) shows that the decay rate is determined by the product of ζ and ω_n. Thus the decay rate is increased by increasing the damping factor or by increasing the system's natural frequency. If we think physically about what's going on with our system, this makes sense. First of all, it certainly seems obvious that increasing the amount of damping in the system will increase the rate of decay, and ζ is linearly proportional to the damping physical damping coefficient, c. But it is also true that a larger ω_n means a higher rate of oscillation, leading to higher velocities and thus more dissipation (since the damping is proportional to velocity). Thus both the damping factor and natural frequency play a part in determining how fast an unforced oscillation will die away.

The general responses we've just examined are absolutely central to understanding and describing the responses of complex systems. The speed at which responses decay and the frequency at which they do so are determining factors in how well the systems will perform the tasks they were designed for.

Example 1.12

Problem How large must c be made for the system described by the following equation to be critically damped?

$$5\ddot{x} + c\dot{x} + 2000x = 0$$

Solution

$$2\zeta\omega_n = \frac{c}{m} = \frac{c}{5}$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{2000}{5}} = 20$$

$$\zeta = \frac{c}{(5)(2)\omega_n} = \frac{c}{200}$$

Since critical damping means $\zeta = 1$ we have

$$c = (200)(1) = 200\,\text{N·s/m}$$

Example 1.13

Problem Consider a system like that shown in earlier Figure 1.26. For this example $m = 10$ kg, $c = 100$ N·s/m, and $k = 30,000$ N/m. If the initial conditions are $x(0) = .01$ m, and $\dot{x}(0) = 0$, what will the response look like for $t > 0$?

Solution Knowing m, c, and k, we can immediately form

$$\omega_n^2 = \frac{30,000}{10}$$

and so

$$\omega_n = 54.8 \text{ rad/s}$$

Knowing that $\frac{c}{m} = 2\zeta\omega_n$ we have

$$\zeta = .0913 \quad \text{and} \quad \omega_d = 54.5 \text{ rad/s}$$

Since

$$x(t) = e^{-\zeta\omega_n t}\left(b_1 \cos(\omega_d t) + b_2 \sin(\omega_d t)\right)$$

we can apply the initial conditions and obtain

$$.01 = b_1$$

and

$$0 = -\zeta\omega_n b_1 + \omega_d b_2$$

Solving both these equations simultaneously gives us

$$b_1 = .01 \quad \text{and}$$

$$b_2 = b_1 \frac{\zeta}{\sqrt{1-\zeta^2}} = .0921$$

Thus

$$x(t) = (.01\cos(54.5t) + .0921\sin(54.5t))\, e^{-5t}$$

1.5 LAGRANGE'S EQUATIONS

No vibrations book is complete anymore without a discussion of Lagrange's equations. All of you have already taken a dynamics class or two, which, in the vast majority of cases, has involved Newtonian mechanics, i.e., finding the equations of motion by applying Newton's laws. The fundamental distinguishing feature of Newtonian mechanics is that it treats the force interactions within a system as the fundamental quantities of interest. Once you've determined all the relevant forces, you simply use $f = ma$ (for constant mass), and you're done. Sounds simple. However, as you probably discovered in class, actually applying these laws is a bit trickier than one might at first think. Correctly analyzing a system using Newton's laws requires us to form free body diagrams and demands that we account for all interaction forces. For instance, let's consider the problem shown in Figure 1.29*a*, that of a mass-pendulum system. The pendulum's base is attached to the mass at A and is free to rotate.

There's more than one way to solve this problem with Newton's laws, and one particular approach to accounting for the applied and interaction forces is shown in Figure 1.29*b*. Note that we *must* solve for the interaction forces (n and t_p) between the two masses, even if we don't really care about them. As vibration engineers, we probably will be satisfied if we have the governing equations of motion, which don't explicitly involve these forces at all.

In the past, the need to spend time getting intermediate results that might be of no interest bothered several researchers, Lagrange being one of them. Lagrange wanted to find an alternate way of finding the equations of motion, one that didn't necessarily involve interaction forces. The notion

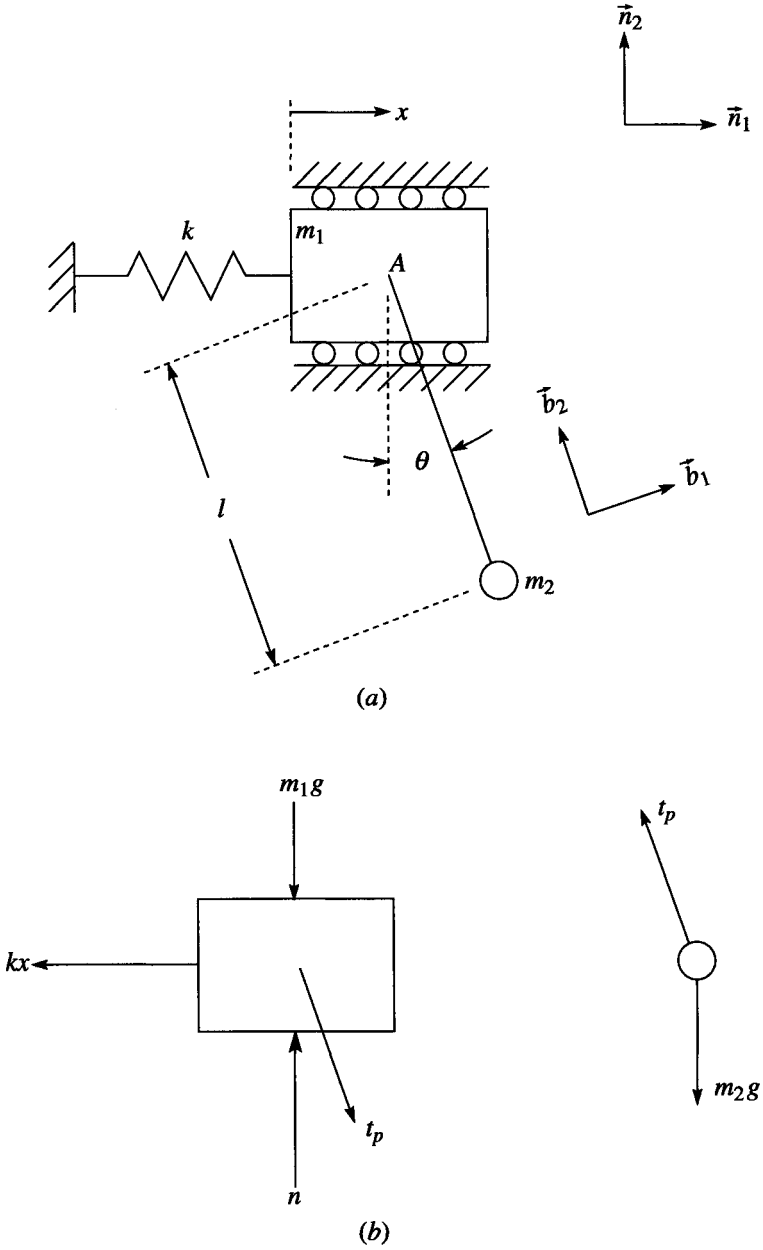


Figure 1.29 Mass/pendulum physical system (*a*) and free body diagram (*b*)

he came up with was to look, not at the forces in a system, but at the *energies*. This viewpoint is quite different from Newton's, and there is a huge body of literature detailing different energy-based approaches to dynamics. Since this isn't a dynamics course, we'll look at only one of these approaches, namely, Lagrange's equations.

It is important at the outset to tell you that we'll be indulging in some major handwaving in the next few pages. The "standard" derivation of Lagrange's equations involves the notion of virtual displacements, something that brings a shudder to some dynamicists, even today. Others couldn't care less. In any event, we'll present this approach, point out the places where one might question it, and leave the rest to you. If you're really interested in getting into the area, there are other, more involved approaches that some people find more appealing.

Okay, enough with the preliminaries. To apply Lagrange's equations, we first need to define what are called *generalized coordinates*. As you know, we can't really do much in the way of dynamics without some coordinates. At the very least, we want our coordinates to exactly pin down the physical configuration of our system, at any point in time. It is very possible for different people to come up with different, and equally valid, sets of generalized coordinates for a given problem. One example is shown in Figure 1.30. Here we have a bar that's supported by two springs. We could decide to use the displacement of the bar at each end as our coordinate set; $x_1(t)$ and $x_2(t)$ are all we need to know to accurately describe the bar's position (we're assuming small motions only). Alternatively, we could use $x(t)$ and $\theta(t)$ to describe the vertical motion of the bar's midpoint and the rotation about this point. This would also work just fine.

Actually, you can come up with any number of coordinate systems; all that matters is that they successfully describe the system's configuration. Most commonly, we refer to these coordinates by the letter q. Thus, for our second set of coordinates in Figure 1.30 we'd have $q_1 = x$ and $q_2 = \theta$.

If we have no externally applied forces and no dissipation, then Lagrange's equations are given by

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \quad i = 1 \cdots n \tag{1.5.1}$$

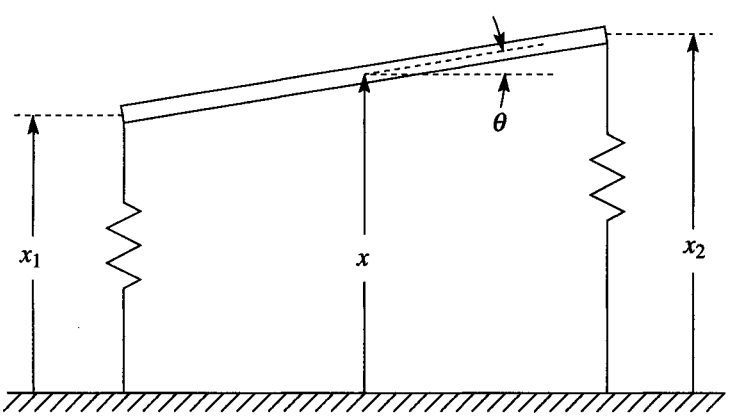


Figure 1.30 Two-coordinate systems

in which

$$L = KE - PE \tag{1.5.2}$$

where KE and PE are the system kinetic and potential energies, respectively, and L is called the Lagrangian.

Where did this result come from? I told you, we weren't going to go into details. You'll just have to believe that it's correct. Or, if you prefer, you can read more about it elsewhere [11]. To demonstrate that it actually works, check out the following example.

Example 1.14

Problem Consider again the system of Figure 1.29. The system consists of a mass m_1 and an attached pendulum. m_1 can only move horizontally, and the pendulum (consisting of a massless rigid link and a lumped mass m_2) is attached to m_1 at the point A. The pendulum is free to rotate. There's no friction in the problem, and no external forces other than gravity. Because gravity will cause the pendulum to oscillate, and the spring k will induce oscillations in m_1, we should expect equations of motion that will involve periodic solutions, presumably ones in which both masses are moving. Find the system's equations of motion.

Solution If we were going to approach this problem from a Newtonian standpoint, we'd first break the system up into two parts and then apply Newton's laws to them individually, as shown in Figure 1.29*b*. The unknown force t_p is the force of interaction between m_1 and m_2; it's what keeps the pendulum attached to m_1. We'll need to set up two force balance equations and one moment balance equation to solve this problem. Note that $\vec{n}_1, \vec{n}_2$ and $\vec{b}_1, \vec{b}_2$ are unit vectors that help define our coordinate systems. The $\vec{b}_1, \vec{b}_2$ frame is fixed to the pendulum while the $\vec{n}_1, \vec{n}_2$ set is fixed to the ground.

The two force balances in the x direction (for m_1 and m_2) are:

$$m_1\ddot{x} = t_p \sin(\theta) - kx$$

and

$$m_2\big(\ddot{x} + l\cos(\theta)\ddot{\theta} - \dot{\theta}^2 l \sin(\theta)\big) = -t_p \sin(\theta)$$

Using these equations together (and eliminating t_p) gives us

$$(m_1 + m_2)\ddot{x} + m_2 l\ddot{\theta}\cos(\theta) - m_2\dot{\theta}^2 l \sin(\theta) + kx = 0$$

Next, we have to sum moments about some convenient point. If we use the pendulum's point of attachment to m_1, our equation will be

$$-m_2 gl \sin(\theta) = m_2\big(l^2\ddot{\theta} + \ddot{x}l\cos(\theta)\big)$$

or, rearranging slightly and canceling out the m_2 term,

$$l^2\ddot{\theta} + \ddot{x}l\cos(\theta) + gl\sin(\theta) = 0$$

which gives us the second equation of motion.

You saw that the interaction force t_p showed up in the initial work and dropped out in the final equations, just as we expected, since we were using a Newtonian approach.

Now let's use Lagrange's equations. To find the kinetic energy, we need to know the velocities of the two masses. Finding m_1's velocity ($\vec{v}_1$) is the easiest, it's just $\dot{x}\vec{n}_1$. m_2 is a bit trickier, since it rotates about m_1. If we use the coordinate vectors shown in Figure 1.29*b*, we see that m_2's velocity is

$$\vec{v}_2 = \dot{x}\vec{n}_1 + l\dot{\theta}\vec{b}_1$$

To find the square of the velocities we need for the kinetic energy, we have to express the velocity in a consistent coordinate system. If we choose the fixed reference frame as the one to use, $\vec{v}_2$ becomes

$$\vec{v}_2 = \left(\dot{x} + l\dot{\theta}\cos(\theta)\right)\vec{n}_1 + l\dot{\theta}\sin(\theta)\vec{n}_2$$

The kinetic energy is simply

$$KE = \tfrac{1}{2}m_1 v_1^2 + \tfrac{1}{2}m_2 v_2^2$$

Using our knowledge of what $\vec{v}_1$ and $\vec{v}_2$ are equal to, we can reexpress this as

$$KE = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\left((\dot{x} + l\dot{\theta}\cos(\theta))^2 + (l\dot{\theta}\sin(\theta))^2\right)$$

The potential energy comes from the translational spring and the gravitational potential that results when the pendulum swings up,

$$PE = \tfrac{1}{2}kx^2 + m_2 gl\,(1 - \cos(\theta))$$

Using these expressions for KE and PE, we can form the Lagrangian ($L = KE - PE$) and apply Lagrange's equations. The two dependent variables (the q_i's) are x and θ. Applying Lagrange's equations with respect to x first, we find

$$\frac{d}{dt}(m_1\dot{x}) + m_2\frac{d}{dt}\left(\dot{x} + \dot{\theta}l\cos(\theta)\right) + kx = 0$$

or, performing the differentiations and grouping terms,

$$(m_1 + m_2)\ddot{x} + m_2 l\ddot{\theta}\cos(\theta) - m_2\dot{\theta}^2 l\sin(\theta) + kx = 0$$

If we next apply Lagrange's equations with respect to θ, we'll obtain

$$m_2\frac{d}{dt}(\dot{x}l\cos(\theta) + l^2\dot{\theta}) - m_2\dot{x}\dot{\theta}l\sin(\theta) + m_2 gl\sin(\theta) = 0$$

Note that the $-m_2\dot{x}\dot{\theta}l\sin(\theta)$ term came about when we differentiated the kinetic energy with respect to θ. Since the kinetic energy has a θ dependence, we obtained this additional term. This almost always happens when you're dealing with rotational systems and rarely occurs for translational ones, since a translational system's kinetic energy usually depends just on $\dot{x}$-type terms, not on terms proportional to x. It's easy to forget this fact if most of your examples are of the translational type, and you might begin to just differentiate the kinetic energy in the $\frac{d}{dt}(\frac{\partial}{\partial\dot{q}})$ term, forgetting about the $\frac{\partial}{\partial q}$ part of the equations. So now you've been warned—make sure to take account of all the relevant differentiations.

We can differentiate and group terms in this last equation to get

$$l^2\ddot{\theta} + \ddot{x}l\cos(\theta) + gl\sin(\theta) = 0$$

If you compare the two sets of equations of motion, you'll quickly notice that they are exactly the same. Thus it appears that Lagrange's equations do what was advertised: they come up with fully valid equations of motion.

Interestingly, you can get into some real battles with certain researchers over whether it is better to use Lagrange's equations or Newton's equations (or even other approaches). However, the simple fact of the matter is that you can almost always get your answer with either approach. In some problems a Newtonian approach might be less work and in others you might find Lagrange's equations to be preferable. It's not a bad idea to simply view the approaches as different tools with which to get the equations of motion and not get too caught up with which is "better." As a general guide, Lagrange's equations are "easier" in that you don't need to use any physical insight in modeling the force interactions. The method simply asks for the energies. Thus you need only find the relevant velocities with which to form the kinetic energy and account for any potential energies. Then you basically plug and chug in Lagrange's equations. The downside is that after you get the velocities, you have to square them and then differentiate. If you've got a bunch of reference frames because your system has several moving parts, then your velocity terms will be long and involved. Therefore you'll have a lot of terms to keep track of. For complicated systems, this can be a big downside, since errors can easily creep in when you're trying to keep track of so many terms. Also, computer simulations will tend to run more slowly because the equations are so complex. But for relatively simple problems, Lagrange's equations are just fine.

Newton's equations shine when you care about interaction forces, since you can't solve the problem without considering them. Often you will care about these interaction terms if you're evaluating the forces in a joint or a slider. The equations are also more physical, since they involve moments and forces, things we're comfortable dealing with as engineers. To apply Newton's equations, you need to find accelerations, something which is generally simpler than the manipulations needed for Lagrange's equations. Also, you need to think physically about the system to construct the necessary free body diagrams, something which isn't necessary with Lagrange's equations. Finally, you have to include all the interaction forces, even if you don't care about them, something that Lagrange's equations don't require.

Hopefully, this short discussion has shown that either approach will work. For the problems we'll be dealing with, most people find Lagrange's equations to be the more convenient. Thus you should probably become familiar with both approaches; it'll help make your life simpler in the long run.

All right, now we've got the unforced, undamped case under control. But what about damping? Can we add a viscous damper to our problem? Yes we can, but it is going to look like a Band-aid kind of fix. If our damper has a known force characteristic, what we need to do is integrate with respect to the time derivative of the generalized coordinates that apply to it. Thus, if we wanted to add a damper to our cart-pendulum problem, we'll get the system shown in Figure 1.31. From our earlier discussions, we recognize that the force of the damper is given by

$$f_d = c\dot{x} \tag{1.5.3}$$

and so, integrating with respect to $\dot{x}$ we obtain

$$\int f_d d(\dot{x}) = \tfrac{1}{2}c\dot{x}^2 \tag{1.5.4}$$

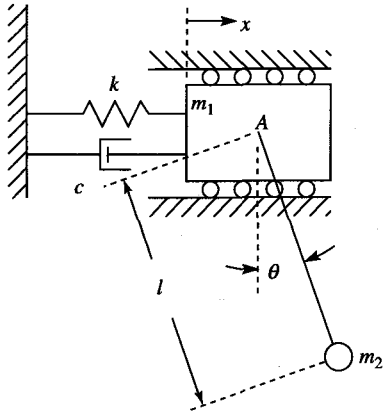


Figure 1.31 Damped mass/pendulum

We define this quantity as the Rayleigh dissipation function, RD. Thus

$$RD \equiv \tfrac{1}{2}c\dot{x}^2 \tag{1.5.5}$$

To use this in Rayleigh's equations, we need to modify the equation a bit, to

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} + \frac{\partial RD}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \qquad i = 1 \cdots n \tag{1.5.6}$$

This modification will allow us to deal with any linear, unforced problem we might encounter.

At this point we have no place left to run. We have to deal with virtual quantities. And just what is a virtual quantity, you might ask? The simplest type is the virtual displacement. The first thing a virtual displacement must satisfy is that it *not* violate any of the physical constraints of our problem. This isn't too bad, in fact it sounds reasonable. If the displacement were allowed to violate physical constraints, like letting bodies pass through each other, it wouldn't stand much chance of being useful to us. Virtual displacements are also taken to be very small changes in the coordinates that occur in zero time. That's right, no time elapses during the virtual change of coordinates.

The following typical constraint equation

$$f(x_1,\ x_2,\ \ldots,\ t) = c \tag{1.5.7}$$

implies that there exists a constraint relation between the coordinates x_i. For instance, if our problem involved the motion of a mass on a table, and x_1 represented displacements to the right or left on the table, x_2 represented displacements forward or back on the table, and x_3 represented displacements above or below the table, our constraint equation would be

$$x_3 = 0 \tag{1.5.8}$$

which tells us that the particle stays on the surface of the table.

If we give our system a virtual displacement then we have

$$f(x_1 + \delta x_1,\ x_2 + \delta x_2,\ \ldots,\ t) = c \tag{1.5.9}$$

This tells us that, even though we've perturbed our system by the virtual displacements δx_1, etc., the constraints are still satisfied. Note that since the virtual displacements don't involve time, t wasn't varied.

We know that our usual idea of work is that force, multiplied by a displacement in the same direction as the force, gives us an associated work. A similar statement is made for virtual work. In this case, a force times a virtual displacement gives us a virtual work.

What we'll do is define the virtual work done on a mass particle as

$$\delta W_i \equiv Q_i \delta q_i \tag{1.5.10}$$

where Q_i is the total force acting on the particle in the direction defined by q_i. This is exactly analogous to our usual definition of work—it's just force times displacement.

Having decided what virtual work is, we can now introduce what, for us, will be the final form of Lagrange's equations,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} + \frac{\partial RD}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i, \quad i = 1 \cdots n \tag{1.5.11}$$

where the Q_i are the virtual forces associated with external forces acting on our system. The use of this equation is best illustrated by an example.

Example 1.15

Problem Figure 1.32 shows the system we've already examined (Figure 1.31) but with the addition of an external force $\vec{f}_{\text{ext}} = f_{\text{ext}}\vec{n}_1$. Determine the Q_i for this problem and form the overall equations of motion.

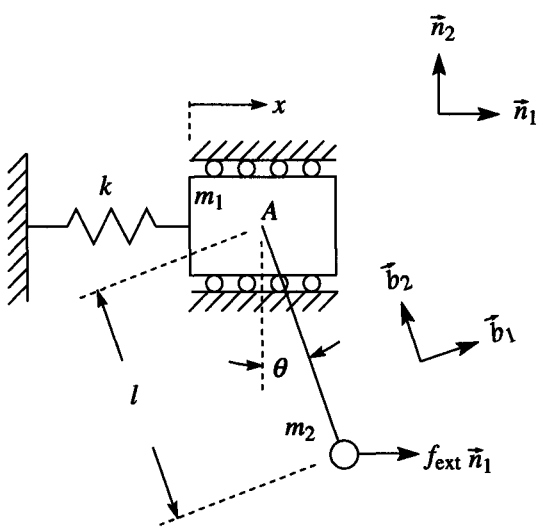


Figure 1.32 Mass/pendulum with external force

Solution The virtual work due to $\vec{f}_{\text{ext}}$ is given by $f_{\text{ext}}\vec{n}_1$ dotted into the virtual displacement of the tip mass. The quickest way to find this quantity is to form the tip mass's velocity, multiply by dt to clear t from the expression, and then switch the dq_i terms to δq_i,

$$\vec{v}_{m_2} = \frac{dx}{dt}\vec{n}_1 + l\frac{d\theta}{dt}\vec{b}_1$$

Multiplying by dt gives

$$d\vec{r}_{m_2} = dx\vec{n}_1 + ld\theta\vec{b}_1$$

and switching to δ yields

$$\delta\vec{r}_{m_2} = \delta x\vec{n}_1 + l\delta\theta\vec{b}_1$$

If we express this purely in terms of the $\vec{n}_i$ coordinate frame we'll see that the virtual displacement is equal to

$$\delta\vec{r}_{m_2} = \delta x\vec{n}_1 + l\delta\theta(\cos(\theta)\vec{n}_1 + \sin(\theta)\vec{n}_2)$$

or, by grouping terms,

$$\delta\vec{r}_{m_2} = (\delta x + l\delta\theta\cos(\theta))\vec{n}_1 + l\delta\theta\sin(\theta)\vec{n}_2$$

The total virtual work is given by

$$f_{\text{ext}}\vec{n}_1 \cdot \big((\delta x + l\delta\theta\cos(\theta))\vec{n}_1 + l\delta\theta\sin(\theta)\vec{n}_2\big) = f_{\text{ext}}\delta x + f_{\text{ext}}l\cos(\theta)\delta\theta$$

If we assign q_1 to be x and q_2 to be θ, then we have

$$Q_1 = f_{\text{ext}}$$
$$Q_2 = l\cos(\theta)f_{\text{ext}}$$
$$(m_1 + m_2)\ddot{x} + m_2 l\ddot{\theta}\cos(\theta) - m_2\dot{\theta}^2 l\sin(\theta) + kx = f_{\text{ext}}$$

and

$$l^2\ddot{\theta} + \ddot{x}l\cos(\theta) + gl\sin(\theta) = l\cos(\theta)f_{\text{ext}}$$

1.6 HOMEWORK PROBLEMS

Section 1.2

1.1. If $x(t) = a_1e^{i\omega t} + a_2e^{-i\omega t}$, $x(0) = 4$ and $\dot{x}(0) = 2$, what are a_1 and a_2 equal to?

1.2. Express $(1 + 2i)e^{i\omega t} + (1 - 2i)e^{-i\omega t}$ in terms of $\sin(\omega t)$ and $\cos(\omega t)$.

1.3. If $x(t) = b_1\cos(\omega t) + b_2\sin(\omega t)$, with the same initial conditions as in Problem 1.1, what are b_1 and b_2 equal to?

1.4. Express $2\cos(3t) + 4\sin(3t)$ in terms of e^{3it} and e^{-3it}.

1.5. Express $\cos^3(\omega t)$ in terms of $\cos(\omega t)$ and $\cos(3\omega t)$ by expressing $\cos(\omega t)$ as $\frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$, cubing the expression, and then grouping the resulting terms.

1.6. Reexpress $\sin(\omega t)\cos^2(\omega t)$ as a sum of harmonic components by expressing $\sin(\omega t)$ and $\cos(\omega t)$ in complex exponential form, carrying through the multiplications and then grouping the terms into real components.

1.7. Express $\sin^2(\omega t)\cos^2(\omega t)$ in terms of $\sin(\omega t)$, $\cos(\omega t)$, $\sin(3\omega t)$, and $\cos(3\omega t)$ by putting the sinusoids in complex exponential form, carrying through the multiplications, and then grouping the resulting terms.

1.8. A sinusoid, $x(t) = a\cos(7t + \phi)$, has initial conditions $x(0) = 3$, $\dot{x}(0) = 20$. Determine a and ϕ.

1.9. Give the position, velocity, and acceleration at $t = 1.5$ if $x(t) = 2\cos(3t) - \sin(3t)$. $x(t)$ is given in meters.

1.10. What is the maximum value of $\dot{x}$ if $x(t) = -\cos(4t) + 2\sin(4t)$? $x(t)$ is given in meters.

1.11. Find the maximum value of $\dot{x}$ if $x(t) = 6e^{-2t} + 2e^{-t}$. $x(t)$ is given in meters.

1.12. A mass moves with the acceleration profile illustrated in Figure P1.12. Determine the position of the mass at $t = 8$ seconds. $x(0) = \dot{x}(0) = 0$.

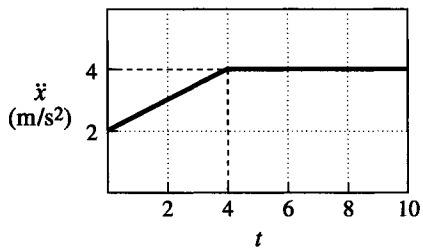


Figure P1.12

1.13. Plot the velocity and displacement vs time graphs for the acceleration profile illustrated in Figure P1.13. $x(0) = 4$ m/s and $x(0) = -2$ m/s.

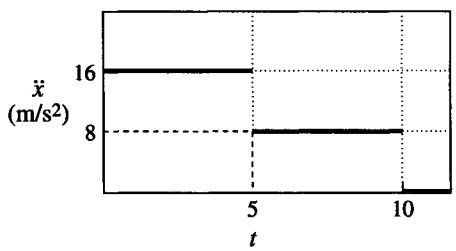


Figure P1.13

1.14. Plot $x(t)$, $\dot{x}(t)$, and $\ddot{x}(t)$, $(0 \le t \le 2\pi)$ for

$$x(t) = 2\cos(t) + .05\sin(10t)$$

What does this tell you about the advisability of differentiating a displacement signal to find the acceleration if the actual signal ($2\cos(t)$ in our case) is corrupted by sensor noise (represented by $.05\sin(10t)$)?

1.15. We know that the spring constant of two identical springs, used in series, is half the individual spring's spring constant. Show from the formula for the stiffness of a coil spring (Appendix B, Figure B.5) that doubling the number of coils (which is equivalent to putting two coil springs in series) will give the same result.

1.16. What is k_{eq} for a serial chains of n springs, each having a spring constant equal to k?

1.17. What is the equivalent stiffness for the set of springs illustrated in Figure P1.17?

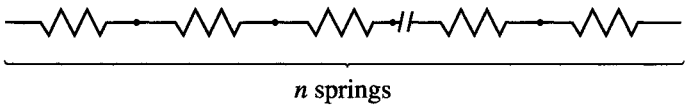


Figure P1.17

1.18. Give the amplitude, frequency, and period of oscillation for the signal illustrated in Figure P1.18.

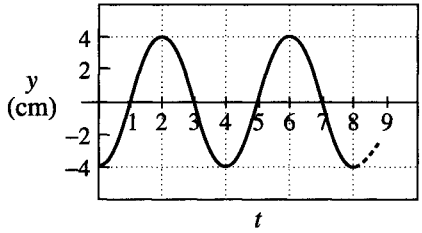


Figure P1.18

1.19. Plot a signal $x(t)$ having a frequency of 2 rad/s and an amplitude of 5 cm for $0 \leq t \leq 10$. Let $x(0) = \frac{5}{\sqrt{2}}$ cm and $\dot{x} = \frac{10}{\sqrt{2}}$ cm/s.

1.20. Determine the phase difference between the two signals illustrated in Figure P1.20. By how much is y lagging x?

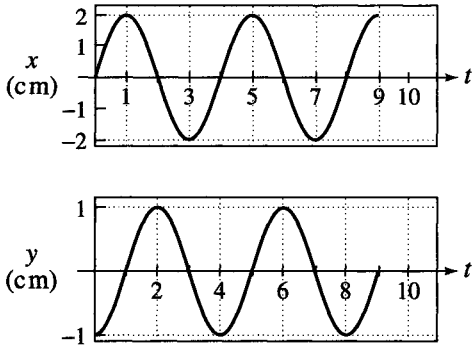


Figure P1.20

1.21. What is ω_n for the system illustrated in Figure P1.21 in terms of m, k_1, k_2, and k_3?

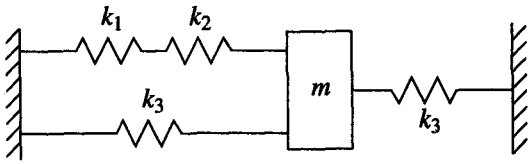


Figure P1.21

1.22. What is ω_n for the system illustrated in Figure P1.22?

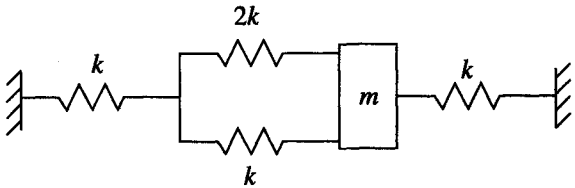


Figure P1.22

1.23. Find the equivalent spring constant for the two springs with widely differing spring constants shown in Figure P1.23. Comment on your result.

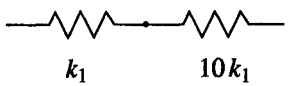


Figure P1.23

1.24. Find the equivalent spring constant for the two springs with widely differing spring constants shown in Figure P1.24. Comment on your result.

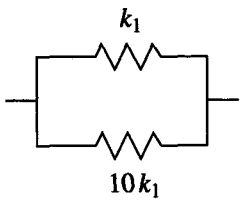


Figure P1.24

1.25. Figure P1.25 shows a Ferris wheel at a carnival. Assume that the sun is directly overhead and that the wheel has a radius of 15 m. It takes 3 minutes to complete a single revolution. Draw a plot of the shadow cast by a single car on the Ferris wheel as a function of time. Identify the amplitude, period, and frequency of the resulting waveform.

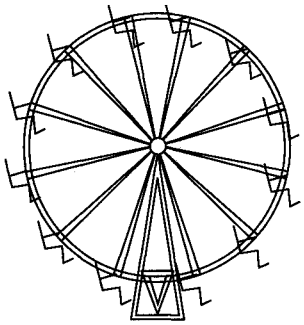


Figure P1.25

1.26. Consider a physical system governed by

$$m\ddot{x} + kx = 0$$

Given some m, k, and initial conditions, is it possible, for the free vibration solution to be given by the following equation?

$$x(t) = 3e^{-2it}$$

1.27. The platform illustrated in Figure P1.27 has a natural frequency of 50 Hz. It is supported by four identical springs. If the platform's mass is equal to 300 kg, determine the spring constant of a single spring. Assume that the platform's displacements are purely vertical.

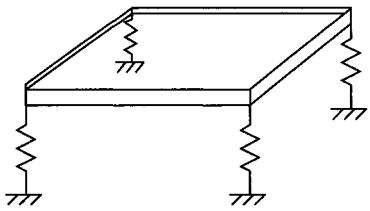


Figure P1.27

1.28. A mass is suspended from a nonlinear spring as in Figure P1.28. Determine the natural frequency of oscillation, ω_n, for small oscillations about the equilibrium position of the spring-mass system. Note that since the spring is not linear, you'll have to linearize about the system's equilibrium position to find its equivalent linear stiffness. The spring's force/displacement relation is

$$f_{\text{spring}} = 2x^3$$

where $f(x)$ is in newtons and x is in meters; $m_1 = 1$ kg and $g = 9.81$ m/s^2.

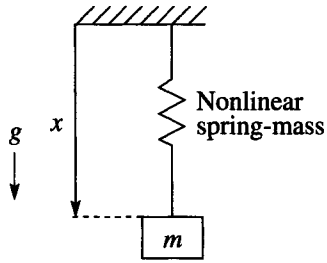


Figure P1.28

1.29. A spring is suspended from a nonlinear spring as in Figure P1.28. The force/displacement relation of the spring is given by

$$f_{\text{spring}} = 49(x + x^2)$$

where f_{spring} is in newtons and x is in centimeters. Find the linearized ω_n for the spring-mass system about its static equilibrium for $m = 10$ kg. Then find the new value of ω_n for $m = 20$ kg. Did the natural frequency drop by 29% (as linear theory would predict)?

1.30. A mass is suspended from a nonlinear spring as in Figure P1.28. The force/displacement relation of the spring is given by

$$f_{\text{spring}} = 10x - 4x^2 + x^3$$

x is measured in centimeters and f_{spring} is given in newtons. Find the linearized ω_n for the system illustrated about its static equilibrium using $m = .4209$ kg. Then find the new value of ω_n for $m = .8418$ kg. Did the natural frequency drop by 29% (as linear theory would predict)?

1.31. Consider the system illustrated in Figure P1.31, which has two linear springs. The unstretched length of k_1 is .5 m and the unstretched length of k_2 is .25 m. $k_1 = 1000$ N/m, $k_2 = 2000$ N/m, $m = 2$ kg, $l = .5$ m. Find the equilibrium position of the mass and determine the natural frequency of the system. Compare this natural frequency to that associated with $l = .75$ m (i.e., no precompression in the spring).

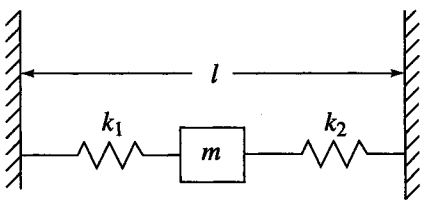


Figure P1.31

1.32. Consider the system in which the linear spring of Problem 1.31 is replaced with a nonlinear spring, as shown in Figure P1.32. The unstretched length of k_1 is .5 m and the unstretched length of the nonlinear spring is .25 m. $k_1 = 1000$ N/m, $f(x) = 24,000x^2$ N, $m = 2$ kg, $l = .5$ m. Find the equilibrium position of the mass and find the nonlinear spring's equivalent linear stiffness at that

deflection. Calculate the natural frequency. Does the natural frequency change if l is set equal to .75 m? How does this situation differ from that of Problem 1.31?

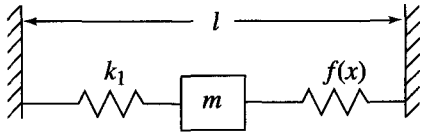


Figure P1.32

1.33. In an effort to add realism to cartoons, you've been asked to dynamically analyze a cartoon gag (assume that you're working during the summer as an animator). In this gag, Dopey Dog is falling from an airplane (see Figure P1.33) and, as he passes a ledge, he grabs an elastic rope he'd tied there earlier. His velocity at the instant he grabs the rope is 31.32 m/s. His mass is 35 kg, and the rope can be considered massless. The spring constant of the rope is 420 N/m. Calculate Dopey's position as a function after grabbing the rope and determine how long it will take for his velocity to reach zero. Assume he doesn't hit the canyon floor. (*Note*: Don't neglect the static offset he'd have due to gravity.) Calculate a trajectory for motions only for the period in which the rope is under tension.

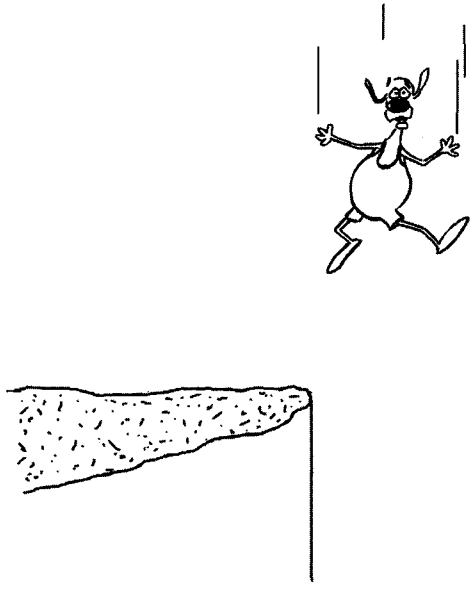


Figure P1.33

1.34. Determine the natural frequencies for the systems illustrated in Figure P1.34. What is the difference between the two cases? Which springs are most active and which are relatively uninvolved in the oscillations?

(*a*) $k_1 = 5\,\text{N/m},\ k_2 = 5\,\text{N/m},\ k_3 = 100\,\text{N/m},\ m = 100\,\text{kg}$

(*b*) $k_1 = 50\,\text{N/m}$, $k_2 = 50\,\text{N/m}$, $k_3 = 10\,\text{N/m}$, $m = 100\,\text{kg}$

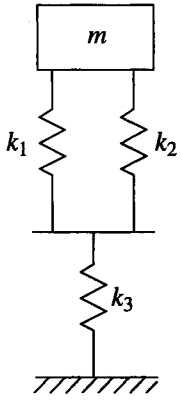


Figure P1.34

1.35. A static horizontal load of 80 N, applied to the lamp housing of the street lamp in Figure P1.35, caused a horizontal deflection of 5 cm. Assume that the motion of an earthquake can be modeled as a step change in position. Thus the initial condition is $x(0) = 18$ cm, $\dot{x}(0) = 0$. Determine the response of the lamp housing. Consider the lamp-pole system to be massless and the mass of the housing to be 25 kg.

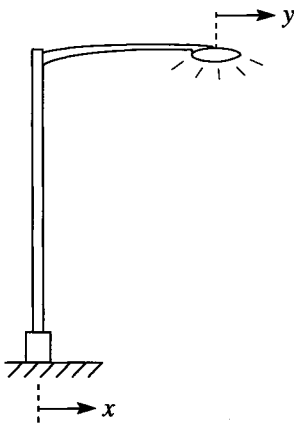


Figure P1.35

1.36. The spring-mass system in Figure P1.36*a* has a natural frequency of ω_1 rad/s. By how much will the natural frequency change if the mass is attached as shown in Figure P1.36*b*? (The spring is cut into two equal pieces and reattached.)

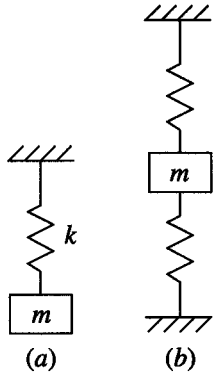


Figure P1.36

1.37. A 56.25 kg mass placed on top of an existing 100 kg mass (part of a spring-mass system) reduces the original natural frequency by 2 rad/s. What is the spring constant k?

1.38. Determine the free vibration of the system in Figure P1.38 (water in a U-shaped tube). The tube has an inner area of s, the water has density ρ, and the length of the water-filled section is l.

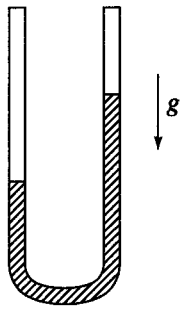


Figure P1.38

1.39. Is there any difference between the systems shown in Figure P1.39? If so, what is it?

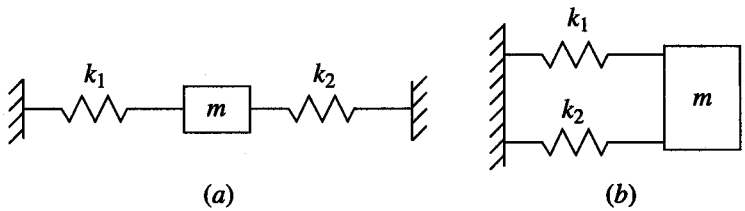


Figure P1.39

1.40. What is the natural frequency of oscillation for the system in Figure P1.40?

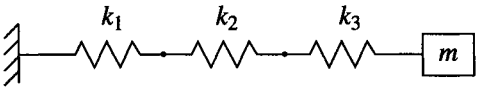
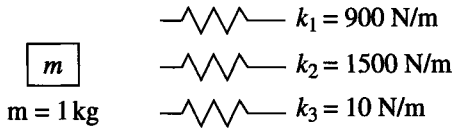


Figure P1.40

1.41. Determine how many natural frequencies you can come up with by combining the given springs in Figure P1.41 with a 1 kg mass.



Some examples

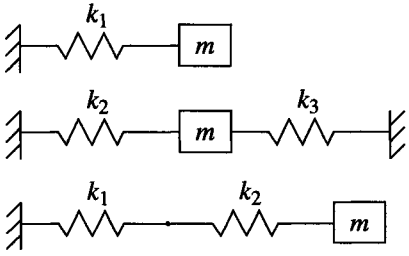


Figure P1.41

1.42. Consider two masses in Figure P1.42 joined by a spring and sliding on a frictionless surface. If the distance between them is given by z, determine the equation of motion of the system that involves only z, m_1, and m_2. (*Note*: The motion will be a "breathing" action in which the masses move together and apart.)

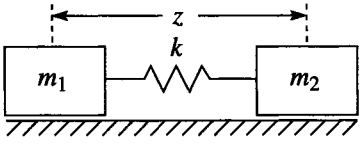


Figure P1.42

1.43. With use, the end conditions of the system in Figure P1.43 have gone from cantilevered-cantilevered to pinned-pinned. What is the percentage change in the system's natural frequency? Consider the beams to be massless. (Use Appendix B.)

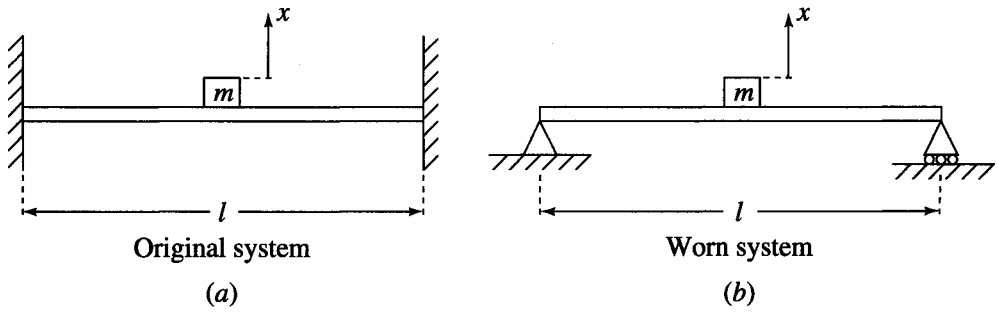


Figure P1.43

1.44. By what percentage will the natural frequency of the original system of Figure P1.43 change if the right support breaks, leaving the end unrestrained? Treat the beam as massless.

1.45. Figure P1.45*a* gives a simple model of a building consisting of a horizontal floor supported by two vertical beams. An even more simplified model would consist of a single beam and mass as shown in Figure P1.45*b*. If the stiffness of each original beam (for which both ends remain vertical) is k, determine what m' must equal in the single-mass, single-beam case if the natural frequencies are to agree. (For the single-mass, single-beam case, the beam end doesn't remain vertical.)

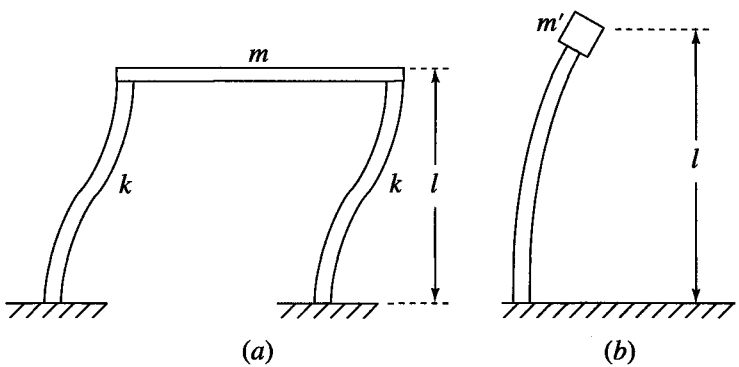


Figure P1.45

1.46. Demonstrate that the stiffness of the system in Figure P1.46*a* is equal to $\frac{192EI}{l^3}$. Use the knowledge that the stiffness of system in Figure P1.46*b* is $\frac{12EI}{l^3}$.

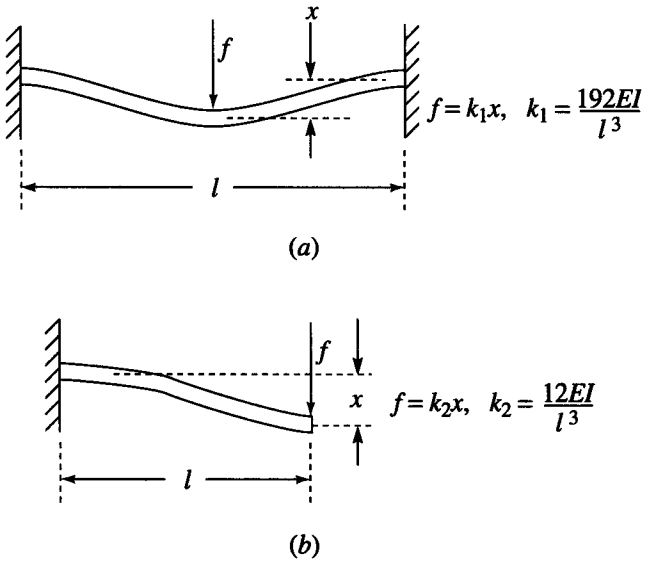


Figure P1.46

1.47. Demonstrate that the stiffness of the system in Figure P1.47*a* is equal to $\frac{48EI}{l^3}$. Use the knowledge that the stiffness of the system in Figure P1.47*b* is $\frac{3EI}{l^3}$.

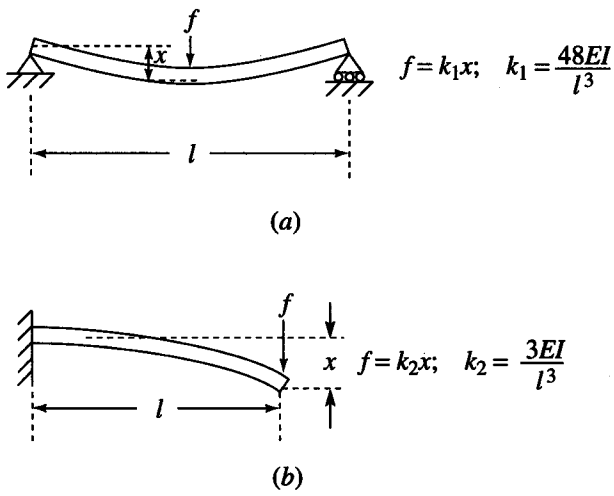


Figure P1.47

1.48. An SDOF spring-mass system has a natural frequency equal to 15 rad/s. How will this change if the suspension's spring constant is doubled?

1.49. When a 1500 kg mass is placed on a suspension, the springs deflect 1 cm. The spring force relation is linear. What is the natural frequency of oscillation for the system?

1.50. What is the natural frequency for the system illustrated in Figure P1.50? (Assume that it oscillates about its equilibrium position.) The radius of each pulley is .3 m, $k = 600$ N/m, and $m = 20$ kg. The pulleys are massless.

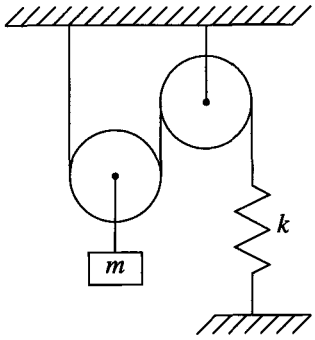


Figure P1.50

1.51. What is the natural frequency for the system illustrated in Figure P1.51? Keep in mind that we're looking at *small* oscillations about equilibrium, so you needn't worry about geometric effects from the springs due to the mass's motion. (*Hint*: Consider using polar coordinates.)

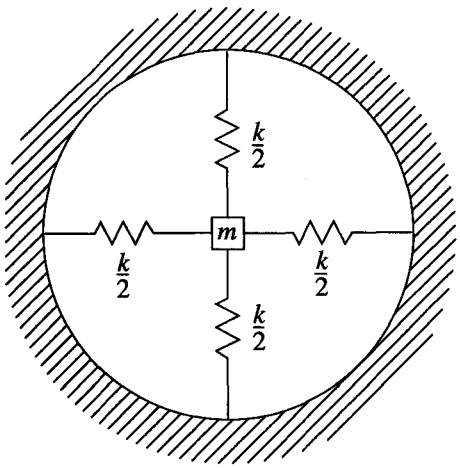


Figure P1.51

1.52. Consider a spring that has an unstretched length of .5 m and a spring constant equal to 500 N/m. Calculate the natural frequency if a 1 kg mass is suspended from the spring. Then calculate how the

natural frequency will change if the spring is cut in half and the mass is suspended from just a half spring.

1.53. The natural frequency of the one-story building shown in Figure P1.53 can be calculated by determining the stiffness of the supporting beams and assuming that the roof moves horizontally. If the width of the building is l_1, the depth is l_2, the stiffness of each beam (4 total) is EI, and the length is l_3, calculate the natural frequency, assuming a roof mass of m. Treat the beams as massless.

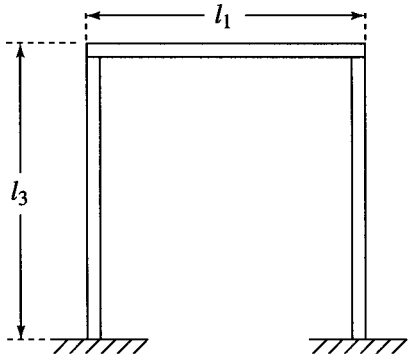


Figure P1.53

1.54. As a practical joke, you've put some fast-acting glue in your friend's slingshot as in Figure P1.54. Your friend puts a mass of .06 kg in the slingshot, pulls it back 1 m, and then releases it. If the mass travels out 1 m beyond the slingshot and then returns to strike his hand .5 second after the release, what must the spring constant be for the massless elastic band? Assume that the elastic band has an unstretched length equal to 0 m.

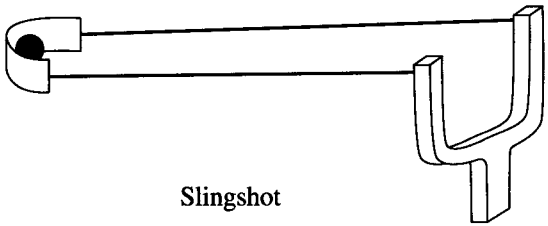


Figure P1.54

1.55. Given the spring force characteristics in Figure P1.55, determine the natural frequency of oscillation of the system. Assume the system executes small motions about the gravitationally induced static equilibrium. Take g to be 10 m/s^2 and $m = 2$ kg.

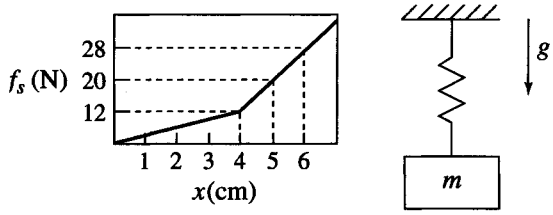


Figure P1.55

1.56. In an effort to add realism to cartoons, you've been asked to dynamically analyze a new episode. Dopey Dog, having attached himself by means of an elastic band to a rocky outcropping (Figure P1.56*a*), is dismayed when the outcropping becomes detached from the cliff (Figure P1.56*b*). After a drop of 49 m, the outcropping lands on a ledge with the part Dopey is attached to projecting outward from the ledge (Figure P1.56*c*). Thus Dopey continues to fall. The unstretched length of the elastic is 1 m. Just before the fall, Dopey was in static equilibrium and was 2 m from the outcropping (i.e., the total length of the unstretched elastic was 2 m). Assume that the separation between Dopey and the outcropping stayed constant during the fall.

Determine the value of linear spring constant k necessary to keep Dopey from hitting the ground (which is 3.1 m below his position at the instant the outcropping hits the second ledge). Considering the actual spring constant (as found from the static equilibrium conditions), will he hit the ground? Finally, which of the assumptions in the foregoing description doesn't make physical sense?

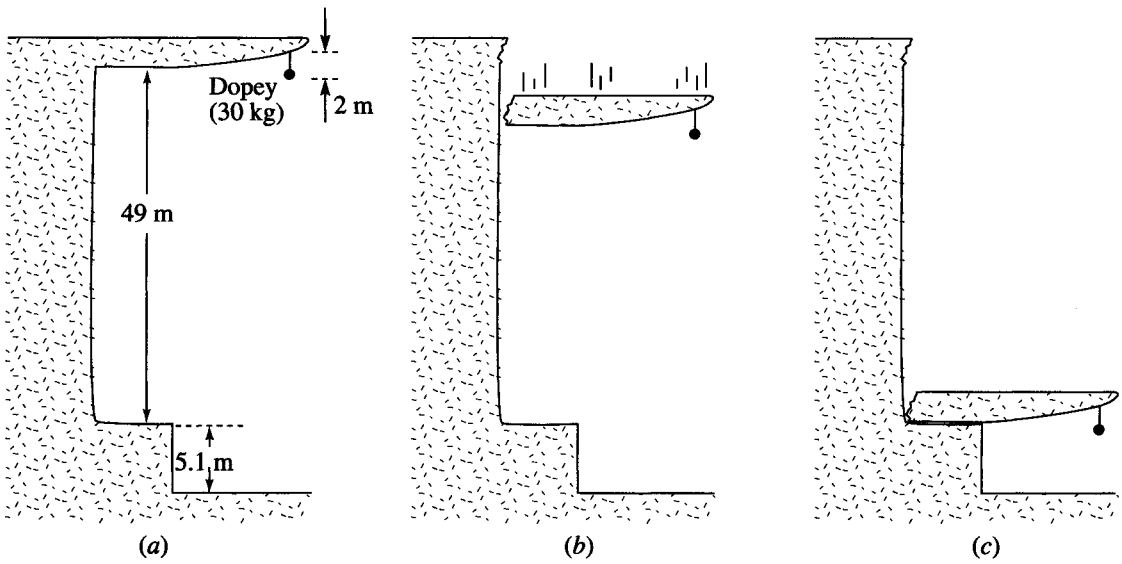


Figure P1.56

1.57. The coupling between the two railroad boxcars in Figure P1.57 can be viewed as a lumped spring. If the masses of the two cars are $m_1 = 2.0 \times 10^5$ kg and $m_2 = .9 \times 10^5$ kg and the spring constant $k = 8 \times 10^5$ N/m, determine the natural frequency of the system's oscillations.

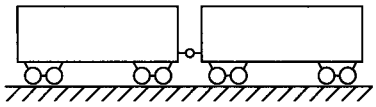


Figure P1.57

1.58. Determine the frequency of oscillation of a cube of wood seen in Figure P1.58 bobbing up and down in the water, in terms of the water's density, the wood's density, g, and l.

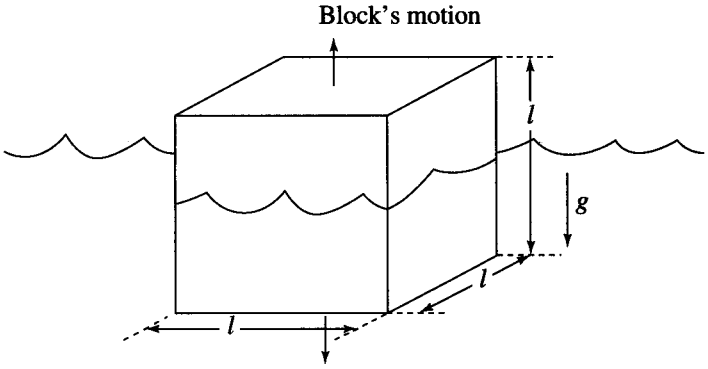


Figure P1.58

Section 1.3

1.59. Use the system illustrated in Figure P1.59 to determine the natural frequency of oscillation about $\theta = 0$. $l = 1.5$ m, $m = 2$ kg, $g = 9.81$ m/s^2.

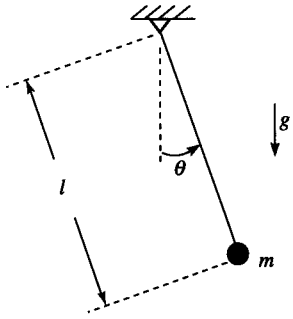


Figure P1.59

1.60. Find the natural frequency about $\theta = 0$ for the system in Figure P1.60. $m_1 = 5$ kg, $m_2 = 5.5$ kg, $l = 1$ m, $g = 9.81$ m/s^2.

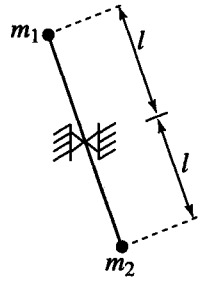


Figure P1.60

1.61. Find the natural frequency about $\theta = 0$ for the system in Figure P.161. $l_1 = 1$ m, $l_2 = 1.1$ m, $m_1 = m_2 = 5$, $g = 9.81$ m/s^2.

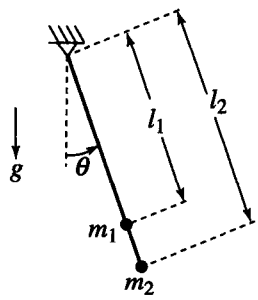


Figure P1.61

1.62. The pendulum illustrated in Figure P1.62 consists of a rigid bar of mass m_2 and length l. Attached to the end is a lumped mass m_1. Find the pendulum's natural frequency of oscillation.

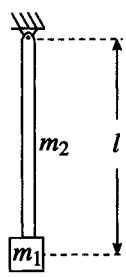


Figure P1.62

1.63. Use the system illustrated in Figure P1.63 to determine the natural frequency of oscillation for the half-disk. Assume no-slip conditions.

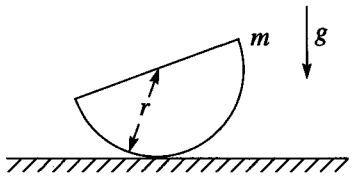


Figure P1.63

1.64. In Figure P1.64 Dopey Dog is rolling a large cylinder up a slope. The actual shape of the ground he's walking on is described by a circle, with a radius of $10r_1$. When $\theta = 30$ degrees, Dopey slips and the cylinder rolls over him. Assuming pure rolling, how long will it be before the cylinder returns to him? The cylinder's mass is m and it has a radius of r_1. Assume that linearization about $\theta = 0$ degrees is a valid approximation.

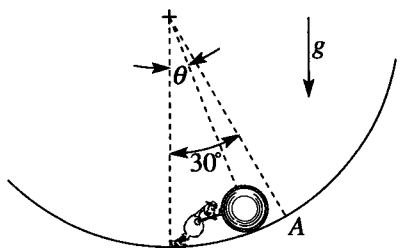


Figure P1.64

1.65. Determine how long it will take the cylinder of Problem 1.64 to return to Dopey if, instead of pure rolling, we have a frictionless interface between the cylinder and the slope.

1.66. For the system shown in Figure P1.66, the mass is constrained to ride in a tube that is rotating in a horizontal plane about the frictionless hinge O at the constant rate ω. The mass is connected to the spring k. Determine the response of the system about O for the following cases:

(a) $\omega^2 < \frac{k}{m}$

(b) $\omega^2 = \frac{k}{m}$

(c) $\omega^2 > \frac{k}{m}$

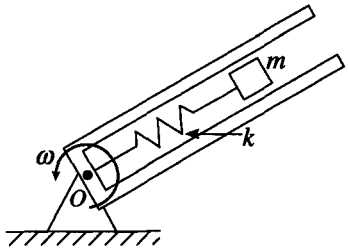


Figure P1.66

1.67. Find the linearized equation of motion about the equilibrium position of the system illustrated in Figure P1.67. You don't have to solve for the equilibrium position itself—just show what you'd need to do to obtain it. The torsional spring is uncompressed when $\theta = 0$. What is the linearized natural frequency?

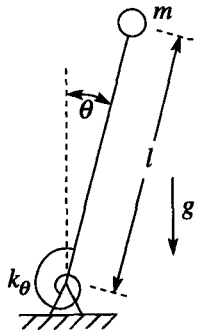


Figure P1.67

1.68. For the system shown in Figure P1.68, find the natural frequency of oscillation for the uniform bar of mass m in the xz plane. The supporting wires are massless and inextensible.

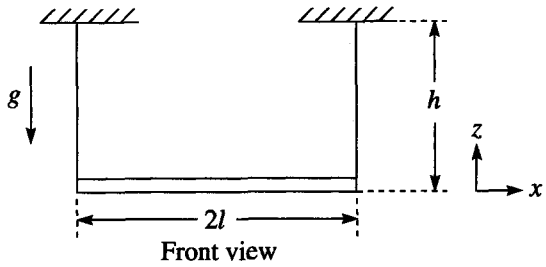


Figure P1.68

1.69. In Figure P1.69 shows a uniform bar (mass m, length $2l$) suspended by two inextensible wires (massless) of length h. At each end of the bar is a lumped mass m_1. What is the frequency of oscillation for rotary oscillations in the x-y plane?

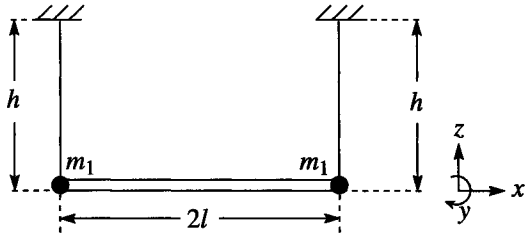


Figure P1.69

1.70. Find the equilibrium position for the system illustrated in Figure P1.70 and compute the natural frequency of oscillations about this equilibrium position. $m_1 = 1$ kg, $l_1 = 1.2$ m, $l_2 = .1$ m, $k = 6000$ N/m.

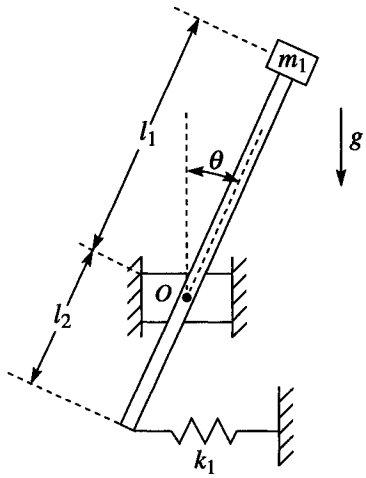


Figure P1.70

1.71. Use Figure P1.71 to determine the natural frequency of oscillation of a point mass sliding without friction on a circular track. (Use a small-angle approximation.)

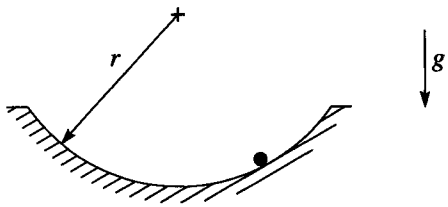


Figure P1.71

1.72. The system illustrated in Figure P1.72 rotates in a horizontal plane. What is the system's natural frequency?

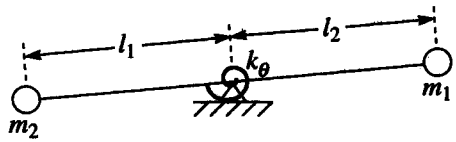


Figure P1.72

1.73. Find the natural frequency of oscillation for the system illustrated in Figure P1.73.

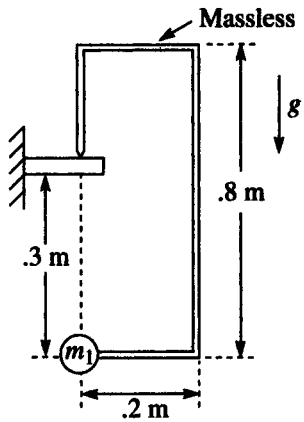


Figure P1.73

1.74. What are the equations of motion for the system shown in Figure P1.74? Consider motions about the vertical position.

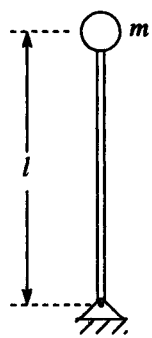


Figure P1.74

1.75. Use Figure P1.75 to find the solution for a mass particle sliding on a semicircle. Consider the system to be frictionless and the equilibrium position to be $\theta = 0$. Linearize the equations of motion (θ small), find

the solutions, and explain what each solution physically represents. (Use $\theta(t) = ae^{\lambda t}$ as the assumed form for the solution.)

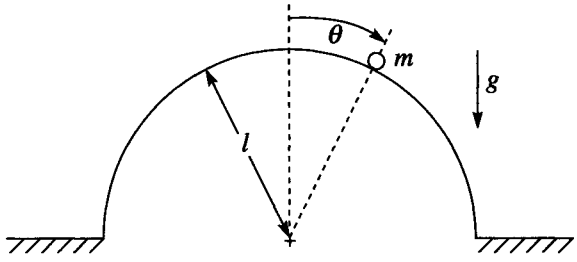


Figure P1.75

1.76. What is the natural frequency for the system shown in Figure P1.76?

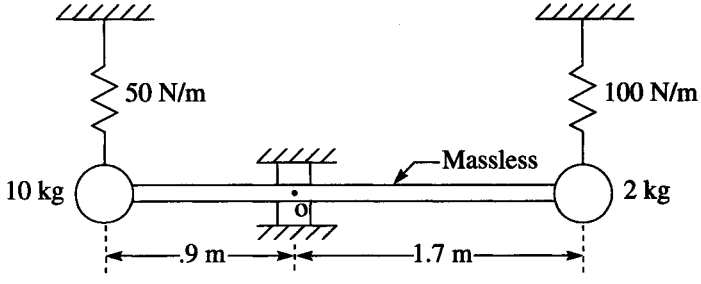


Figure P1.76

1.77. What is the natural frequency for the system shown in Figure P1.77?

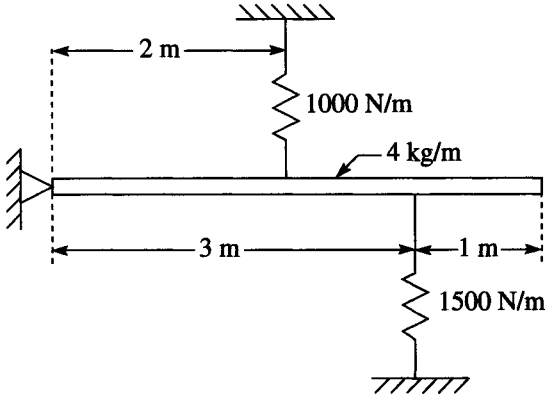


Figure P1.77

1.78. What is the natural frequency for the system shown in Figure P1.78? The hinge at O is frictionless; $k = 10,000$ N/m, $m_1 = 5$ kg, $m_2 = 7$ kg, $\rho = 1$ kg/m, $l_1 = 3$ m, and $l_2 = 5$ m.

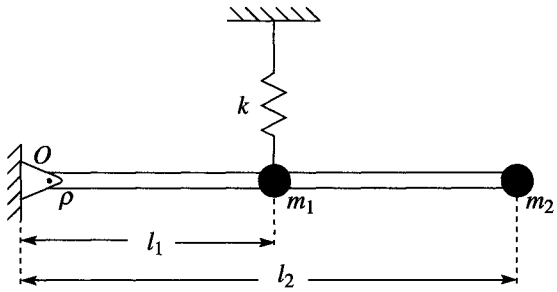


Figure P1.78

1.79. The system shown in Figure P1.79 consists of a mass m that slides without friction along a horizontal track and a tensioned string that connects the mass to the point 0. Determine the natural frequency of the system under the assumption that the tension doesn't change as the mass changes position. $m = 1$ kg, $\rho = .2$ kg/m, $T = 100$ N.

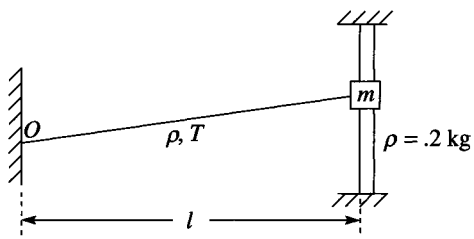


Figure P1.79

1.80. A band saw can be modeled as a continuous loop that runs around two pulleys, as shown in Figure P1.80. The steel has some flexibility, represented by the springs in the figure. Neglect the mass of the steel loop and determine the natural frequency of the pulley/band. Assume that the oscillation mode involves the pulleys moving in opposition to each other. The rotational inertia of each pulley about its shaft is equal to .001 kg·m^2, the spring stiffness is 10,000 N/m, and the radius of the pulleys is .4 cm.

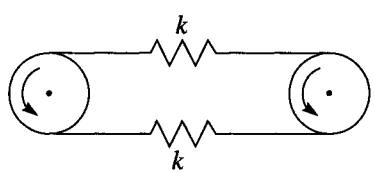


Figure P1.80

1.81. To analyze the motions of an automobile on its tires, we approximate the auto by a bar of uniform mass (1000 kg) with mass moment of inertia equal to 1, 333.3 kg·m^2. The bar is 4 m long. The tire-suspension

assembly is approximated by a pair of springs, with spring constant equal to 11,697 N/m for each one. Use the system illustrated in Figure P1.81 to consider motions in which the car oscillates about its center of mass (no translation). What is the natural frequency of motion for the system?

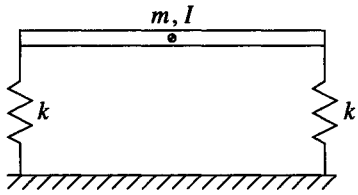


Figure P1.81

1.82. One way to determine a body's mass moment of inertia, $I_{\overline{AO}}$, is to attach the body to a torsional spring, observe the free vibrations, and deduce $I_{\overline{AO}}$ from the frequency of the oscillation. If the torsional stiffness of the vertical rod in Figure P1.82 is 81 N·m/rad and the free vibrations have a frequency of 150 rad/s, what must $I_{\overline{AO}}$ be equal to?

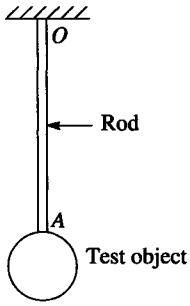


Figure P1.82

1.83. Determine the torsional stiffness of the vertical rod in the system illustrated in Figure P1.83. The rod is rigidly attached to the ceiling at 0 and attached to a bar of length 1 m and mass 3 kg (as shown). If the free vibrations occur at 5 rad/s, what is the rod's torsional stiffness?

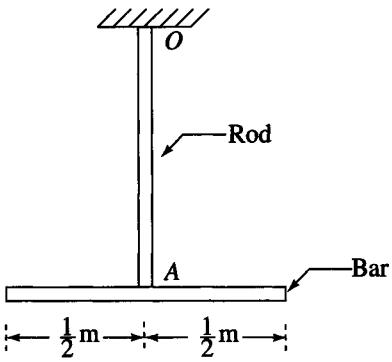


Figure P1.83

1.84. Figure P1.84 shows a lumped mass between two tensioned strings. Treat the strings as massless and neglect gravity. Determine the period of the free vibrations if the tension in the string is equal to 400 N, $l = 2$ m, and the mass is equal to .1 kg. (*Note*: Only small-amplitude motions of the mass take place.)

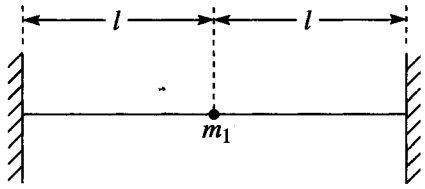


Figure P1.84

1.85. A circular disk is supported by three massless wires, as shown in Figure P1.85. The disk has a density ρ. Determine the frequency of oscillation if the disk is given an initial twist and released. Assume that the initial angular displacement of the disk is small.

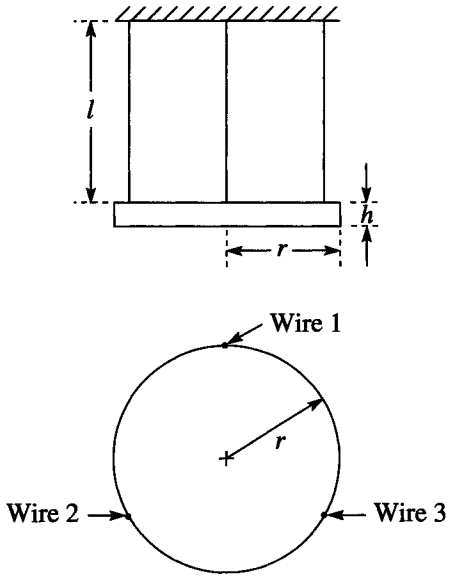


Figure P1.85

1.86. Find ω_n for the system illustrated in Figure P1.86. Consider the beams to be massless. The thin hoop of mass m and radius r can only rotate about O.

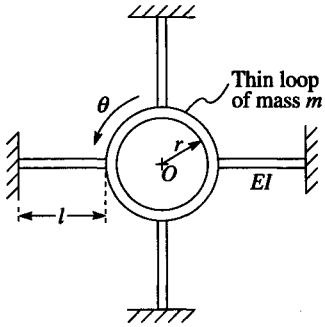


Figure P1.86

1.87. Find ω_n for rotational motions of the system illustrated in Figure P1.87 (a uniform disk constrained by four wires). Consider the tensioned restraining wires to be massless and to have uniform tensions T. All wires are of length l.

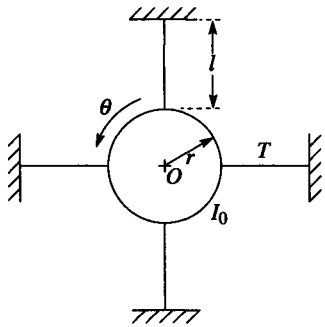


Figure P1.87

1.88. Determine the period of oscillation for the system illustrated in Figure P1.88. (Consider small oscillations, about $\theta = 0$.) $m_1 = 1$ kg, $m_2 = 5$ kg, $l_1 = 2.4$ m, $l_2 = 4$ m.

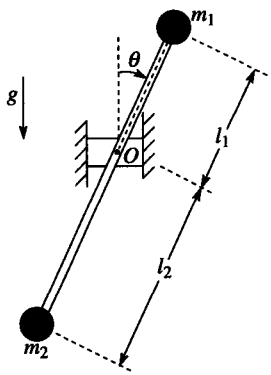


Figure P1.88

Section 1.4

1.89. As long as an SDOF spring-mass-damper is underdamped, its vibrational response will cross zero an infinite number of times, just as in the undamped case. By how much will the "period" of the oscillations be decreased from that of the undamped case if $\zeta = .05$? What if $\zeta = .8$? Comment on the results.

1.90. Does the addition of damping increase or decrease the frequency of the system oscillations?

1.91. If we wish to increase the decay rate for a viscously damped SDOF, is it more effective to alter ω_n or ζ?

1.92. What effect on the decay rate does m have for a viscously damped SDOF system?

1.93. Can the frequency of oscillation for a viscously damped spring-mass-damper system be altered without changing the oscillation decay rate by varying the system parameters?

1.94. If the mass of a spring-mass-damper system is doubled in value while the spring stiffness is halved, how will the decay rate and frequency of subsequent oscillations be affected?

1.95. How many oscillation cycles will elapse if you're waiting for the oscillation amplitude of a spring-mass-damper system to drop by 25% and $\zeta = .04$? Let $\omega_n = 100$. What if ζ is decreased to .0004?

1.96. Oscillation amplitudes are reduced by 50% in one second for a given spring-mass-damper system. Say you want to have the amplitude reduced to 25% of the original value in one second. How effective would it be to vary the value of k?

1.97. What does simultaneously doubling the mass and stiffness of a spring-mass-damper system do to the free vibration decay rate?

1.98. The mass illustrated in Figure P1.98 vibrates in a viscous flow. The force experienced by the mass as a function of velocity is plotted on the graph also shown in the Figure P1.98. Determine the damping factor for this system for $k = 1058$ N/m and $m = 2$ kg.

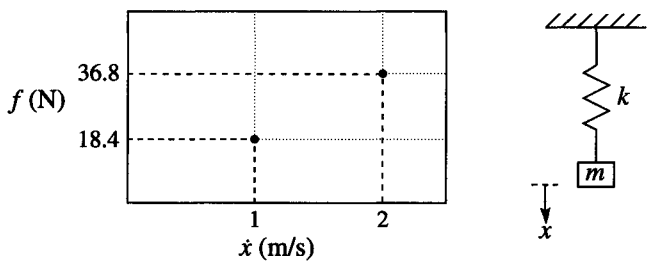


Figure P1.98

Section 1.5

(*Note*: Use Lagrange's equations to find the equations of motion for all the problems in this section.)

1.99. Find the equation of motion for the system illustrated in Figure P1.99.

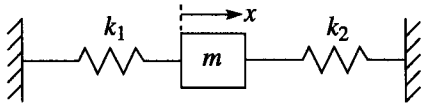


Figure P1.99

1.100. Find the equation of motion for the system illustrated in Figure P1.100.

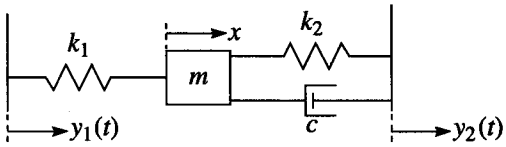


Figure P1.100

1.101. Find the equation of motion for the system illustrated in Figure P1.101.

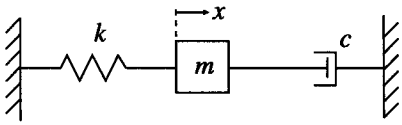


Figure P1.101

1.102. Find the equation of motion for the system illustrated in Figure P1.102.

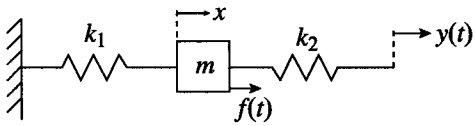


Figure P1.102

1.103. Find the kinetic and potential energies for the system illustrated in Figure P1.103 and determine the full equations of motion. After you've determined the equations of motion, linearize them (retaining terms like x, $\dot{x}$, $\dot{\theta}$, etc.).

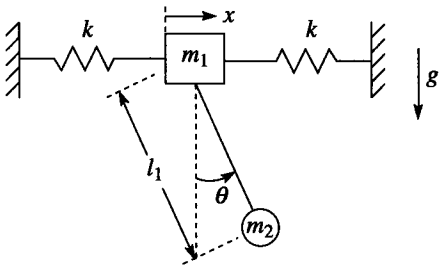


Figure P1.103

1.104. What are the equations of motion for the system illustrated in Figure P1.104?

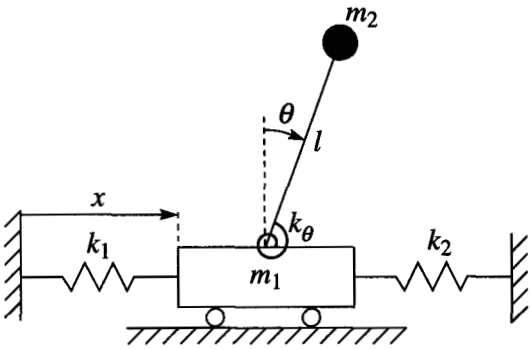


Figure P1.104

1.105. Find the equations of motion for the system illustrated in Figure P1.105.

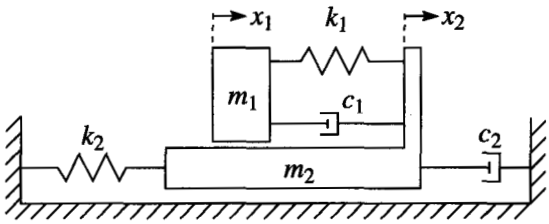


Figure P1.105

1.106. Find the equation of motion for the system illustrated in Figure P1.106 and linearize it about $\theta = 0$.

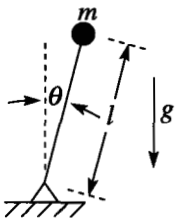


Figure P1.106

1.107. Find the equations of motion for the system illustrated in Figure P1.107.

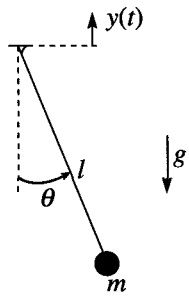


Figure P1.107

1.108. Find the linearized equation of motion for the system illustrated in Figure P1.108. Assume that the cylinder is of mass m and rolls without slipping on the circular path.

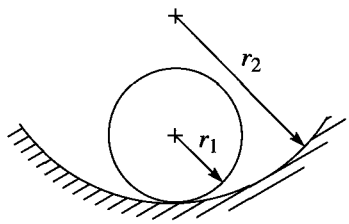


Figure P1.108

1.109. Find the equation of motion for the system illustrated in Figure P1.109. There is no interface friction.

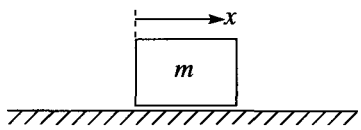


Figure P1.109

1.110. Find the equation of motion for the system illustrated in Figure P1.110. Assume that dry friction ($f = \mu W$) acts between the mass and the supporting surface.

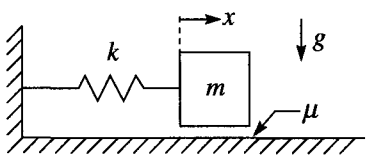


Figure P1.110

1.111. Find the equation of motion for the system illustrated in Figure P1.111. There is no interface friction.

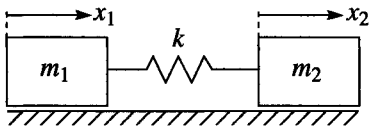


Figure P1.111

1.112. Find and linearize the equations of motion for the system illustrated in Figure P1.112. The bar has length l.

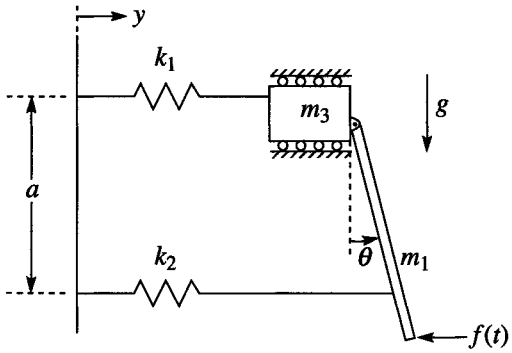


Figure P1.112

1.113. Find the equations of motion for the system illustrated in Figure P1.113. Assume small rotations and vertical deflections of the bar (no motion in the bar's longitudinal direction). Assume that the bar is uniform. As coordinates, use the vertical deflection of the bar's center of mass (y) and the rotation of the bar (θ).

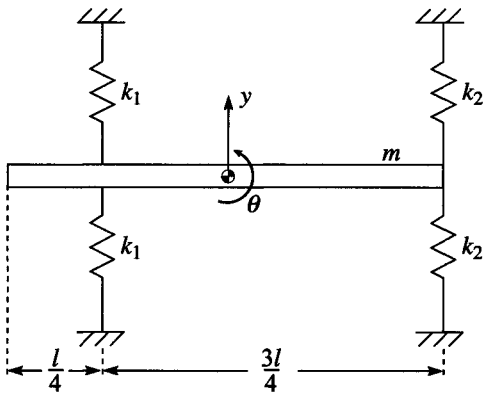


Figure P1.113

1.114. Find and linearize the equations of motion for the system illustrated in Figure P1.114. $m_1 = 1$ kg, $m_2 = 3$ kg, $l_1 = 2$ m, $l_2 = 1$ m, $g = 9.81$ m/s^2.

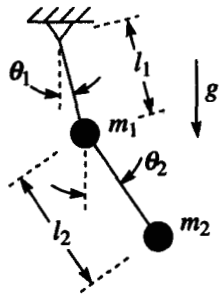


Figure P1.114